TMS320DM270

Digital Media Processor

Data Sheet   Rev β 0.8C

('03/07/07)

Preliminary

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HISTORY

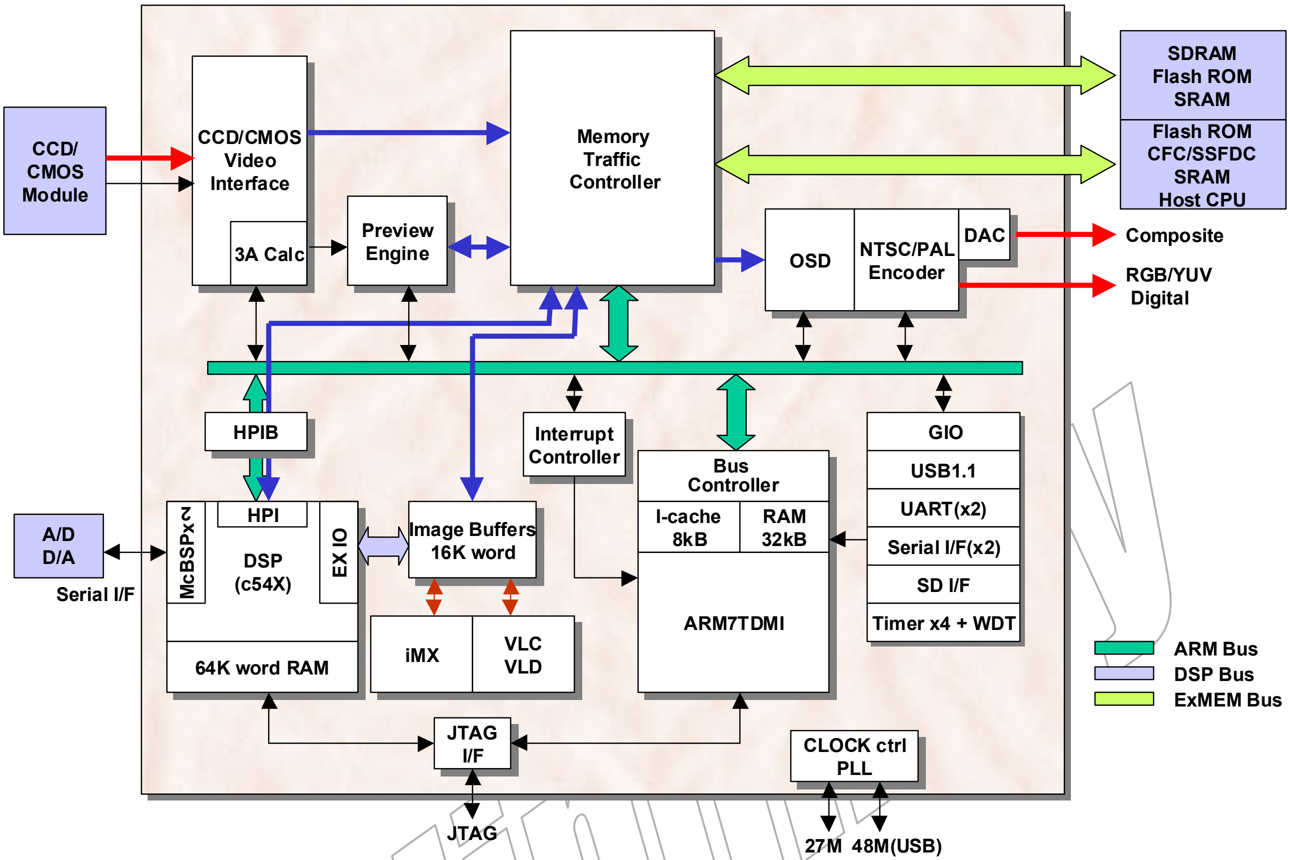
Revision	Date	Notes
β 0.0	25 Jul 02	Initial revision.
β 0.1	29 Aug 02	Modified Functional Block Diagram Modified Pin List (Pin#113,114,112,243)
β 0.2	2 Sep 02	Add table of operating conditions for D/A
β 0.3	11 Sep 02	Modified Pin Direction of DSP_BDR0
β 0.4	24 Sep 02	Modified Ball Number of PLLVDD1 and PLLGND1
β 0.5	04 Dec 02	Revised pin list. - Changed group name Power to GND about xGNDx (core, I/O). - Add pull-dn of SDR_DQx as note *2). - Changed IO type - GIO33(#Pin=16) - EXT_LINE_ID(NTSC/PAL Encoder) : O → I - GIO26(#Pin=25) - C_WEN(CCD In Write Enable) : O → I - GIO22(#Pin=29) - SDEN1(Serial Data Enable 1) : I → I/O - GIO21(#Pin=30) - SCLK1(Serial Clock 1) : I → I/O - GIO13(#Pin=40) - DSP_BDX1(DSP Serial-port Transmit 1) : I/O → O - GIO12(#Pin=41) - DSP_BCLKX1(DSP Transmit Clock 1) : I → I/O - GIO11(#Pin=42) - DSP_BFSX1(DSP Frame Sync Pulse Transmit 1) : I → I/O - GIO10(#Pin=43) - DSP_BDR1(DSP Serial Data Receive 1) : I → I/O - GIO9 (#Pin=44) - DSP_BCLKR1(DSP Receive Clock 1) : I → I/O - GIO8 (#Pin=45) - DSP_BFSR1(DSP Frame Sync Pulse Receive 1) : I → I/O - M48XI(#Pin=117) : I → I/O - MXI (#Pin=231) : I → I/O - EM_WAIT(#Pin=223) : O → I/O - EM_CE0 (#Pin=226) : O → I/O Updated AC timings.
β 0.6	31 Jan 03	"3. Parameter Information" - Changed voltage range (all VDD) in all recommended operating conditions. "4.1 OSC Interface and Clock Timing" - Corrected missing comments. - Changed tc(SDR) from 120MHz into 100MHz. - Changed tw(RESET) from 10cycle(tc(ARM)) into 10cycle(tcMXI))
β 0.7	28 Feb 03	"3. Parameter Information" - Changed voltage range (all VDD) in recommended operating conditions. "4. TIMING DIAGRAMS" - Add preliminary AC data in all sections. - "4.1 OSC Interface and Clock Timing" - Changed tc(ARM) from 90MHz into 88MHz. - Changed tc(SDR) from 100MHz into 102MHz. - Changed tc(DSP) from 100MHz into 102MHz.
β 0.8	7 Jul 03	"2.1 Pin List" - Corrected the Pin Function descriptions for "DSP_xxx(#pin=1-6,8)" - Corrected Pin Function, due to changing I2C module name into Pseudo-I2C - GIO32(#Pin=17) - SDA : I2C Data → Pseudo-I2C Data - GIO31(#Pin=18) - SCL : I2C Clock → Pseudo-I2C Clock - Changed IO type - GIO32(#Pin=17) - SDA(Pseudo-I2C Data) : I → I/O - Changed "GIO0(#pin=53)" function - Pin Name : GIO0 → WAKEUP - IO type : I/O → I "3.2 Recommended operating conditions (DVDD>CVDD)" - Changed the minimum voltage for DVDD : 3.00 → 2.70 - Changed the minimum temperature for Tc : 0 → -20 - Corrected *3) in Note "4.1 OSC Interface and Clock Timing" - Corrected the duplicated descriptions in Note. - Corrected "tw(RESET)" : 10cycle(tc(ARM)) → 10cycle(tc(MXI)) - Added the special condition for "tc(SDR)"

		"4.4 SDRAM Interface timing" - Added the special conditions in Note - "Burst Interrupt by Precharge (BL=4)" - tRC : 6 → 7 - tRAS : 4 → 5 - tRDL : 2 → 3 - Auto Refresh - tRC : 6 → 7 "4.6 Wakrup, General IO (Interrupt Input) timing" - Corrected the number of bit : GIO[7:0] → GIO[15:1] - Added "WAKEUP" timing "4.11 External Host Interface timing" - "External Host Read Timing (DM270 → External Host)" - tRC : 6 → 7 - tRAS : 4 → 5 "4.12 Pseudo-I2C BUS timing" - Changed module name into "Pseudo-I2C" "5. DAA15 Cell" - Added the special conditions in Note

CONTENTS

1. Introduction

1.1 Functional Block Diagram



2. Pin Assignments

2.1 Pin List

PIN #	CSP BALL #	Pin Name	Group	Type	During reset Nrml	During reset Ehost	Pin Function							Pull up/ down
1	B2	DSP_BDX0	Audio I/F	O	Z	Z	**DSP Serial-port Transmit 0**							-
2	B1	DSP_BCLKX0	Audio I/F	I/O	I	I	**DSP Transmit Clock 0**							-
3	E5	DSP_BFSX0	Audio I/F	I/O	I	I	**DSP Frame Sync Pulse Transmit 0**							-
4	C2	DSP_BDR0	Audio I/F	I/O	I	I	**DSP Serial Data Receive 0**							-
5	D3	DSP_BCLKR0	Audio I/F	I/O	I	I	**DSP Receive Clock 0**							-
6	F6	DSP_BFSR0	Audio I/F	I/O	I	I	**DSP Frame Sync Pulse Receive 0**							-
7	C1	DVDD1	Power	-	-	-	VDD for I/O							-
8	D2	DSP_CLKS0	Audio I/F	I	I	I	**DSP Serial Clock 0**							-
9	H8	DGND1	GND	-	-	-	GND for I/O							-
10	E3	RXD0	Communication I/F	I	I	I	UART RXD 0 / AIMSIF Data in							-
11	F5	TXD0	Communication I/F	O	H	H	UART TXD 0 / AIMSIF Data out							-
12	D1	SDEN0	Communication I/F	I/O	I	I	Serial Data Enable 0 / AIMSIF Enable							-
13	E2	SCLK0	Communication I/F	I/O	I	I	Serial Clock 0 / AIMSIF Clock							-
14	E1	SDI0	Communication I/F	I	I	I	Serial Data input 0							-
15	F3	SDO0	Communication I/F	O	L	L	Serial Data Output 0							-
16	F1	GIO33	General I/O	I/O	I	I	GIO	EXT_LINE_ID	I	NTSC/PAL Encoder				-
17	G5	GIO32	General I/O	I/O	I	I	GIO	SDA	**I/O**	Pseudo-I2C Data				-
18	F2	GIO31	General I/O	I/O	I	I	GIO	SCL	O	Pseudo-I2C Clock				-
19	H7	GIO30	General I/O	I/O	I	I	GIO	PWM1	O	Pulse Width Modulation				-
20	G3	GIO29	General I/O	I/O	I	I	GIO	PWM0	O	Pulse Width Modulation				-
21	G1	CVDD1	Power	-	-	-	VDD for core							-
22	H10	CGND1	GND	-	-	-	GND for core							-
23	H5	GIO28	General I/O	I/O	I	I	GIO	FIELD	O	Field (NTSC/PAL Enc.)				-
24	G2	GIO27	General I/O	I/O	I	I	GIO	CCDC_SHUT	O	CCD CCDC_SHUT				-
25	J8	GIO26	General I/O	I/O	I	I	GIO	C_WEN	I	CCD In Write Enable				-
26	H3	GIO25	General I/O	I/O	I	I	GIO	MIRQ	O	IRQ Monitor	MRST	O	Master Reset from WDT	-
27	H1	GIO24	General I/O	I/O	I	I	GIO	TXD1	O	UART TXD 1				-
28	H2	GIO23	General I/O	I/O	I	I	GIO	RXD1	I	UART RXD 1				-
29	J6	GIO22	General I/O	I/O	I	I	GIO	SDEN1	I/O	Serial Data Enable 1				-
30	J5	GIO21	General I/O	I/O	I	I	GIO	SCLK1	I/O	Serial Clock 1				-
31	J3	GIO20	General I/O	I/O	I	I	GIO	SDI1	I	Serial Data input 1				-
32	J1	GIO19	General I/O	I/O	I	I	GIO	SDO1	O	Serial Data Output 1				-
33	J2	GIO18	General I/O	I/O	I	I	GIO	CLKOUT2	O	General Clock Out 2				-
34	J7	GIO17	General I/O	I/O	I	I	GIO	CLKOUT1	O	General Clock Out 1				-
35	K5	GIO16	General I/O	I/O	I	I	GIO	CLKOUT0	O	General Clock Out 0				-
36	K3	GIO15	General I/O	I/O	I	I	GIO	INT15	I	External Interrupt	DSP_XF	O	DSP External Flag	-
37	K1	DVDD2	Power	-	-	-	VDD for I/O							-
38	K2	GIO14	General I/O	I/O	I	I	GIO	INT14	I	External Interrupt	DSP_CLKS1	I	DSP Serial Clock 1	-
39	H11	DGND2	GND	-	-	-	GND for I/O							-
40	K7	GIO13	General I/O	I/O	I	I	GIO	INT13	I	External Interrupt	DSP_BDX1	O	DSP Serial-port Transmit 1	-
41	L1	GIO12	General I/O	I/O	I	I	GIO	INT12	I	External Interrupt	DSP_BCLKX1	I/O	DSP Transmit Clock 1	-
42	L3	GIO11	General I/O	I/O	I	I	GIO	INT11	I	External Interrupt	DSP_BFSX1	I/O	DSP Frame Sync Pulse Transmit 1	-
43	L5	GIO10	General I/O	I/O	I	I	GIO	INT10	I	External Interrupt	DSP_BDR1	I/O	DSP Serial Data Receive 1	-
44	L2	GIO9	General I/O	I/O	I	I	GIO	INT9	I	External Interrupt	DSP_BCLKR1	I/O	DSP Receive Clock 1	-
45	L6	GIO8	General I/O	I/O	I	I	GIO	INT8	I	External Interrupt	DSP_BFSR1	I/O	DSP Frame Sync Pulse Receive 1	-
46	M1	GIO7	General I/O	I/O	I	I	GIO	INT7	I	External Interrupt				-
47	M3	GIO6	General I/O	I/O	I	I	GIO	INT6	I	External Interrupt				-
48	M2	GIO5	General I/O	I/O	I	I	GIO	INT5	I	External Interrupt				-
49	L7	GIO4	General I/O	I/O	I	I	GIO	INT4	I	External Interrupt				-
50	N1	GIO3	General I/O	I/O	I	I	GIO	INT3	I	External Interrupt				-
51	M5	GIO2	General I/O	I/O	I	I	GIO	INT2	I	External Interrupt				-
52	N3	GIO1	General I/O	I/O	I	I	GIO	INT1	I	External Interrupt				-
53	N2	**WAKEUP**	**Wakeup**	**I**	**I**	**I**								-
54	P1	CVDD2	Power	-	-	-	VDD for core							-
55	L8	RSN	Misc.	I	I	I	Reset for Main Land							-
56	P2	PWDWN	Sleep Controller	I	I	I	Power Down Mode							Dn

PIN #	CSP BALL #	Pin Name	Group	Type	During reset Nrml	During reset Ehost	Pin Function	Pull up/ down
57	H12	CGND2	GND	-	-	-	GND for core	-
58	P3	TCK	Misc.	I	I	I	JTAG Clock	Up
59	N5	TDI	Misc.	I	I	I	JTAG Data Input	Up
60	R1	TMS	Misc.	I	I	I	JTAG Test Mode Select	Up
61	M7	TRST	Misc.	I	I	I	JTAG Test Reset	Dn
62	R2	TEST0	Misc.	I	I	I	Test Input0	Dn
63	R3	TEST1	Misc.	I	I	I	Test Input1	Dn
64	T1	TEST2	Misc.	I	I	I	Test Input2	Dn
65	T2	TEST3	Misc.	I	I	I	Test Input3	Dn
66	U1	SCANEN	Misc.	I	I	I	Test Scan Enable	Dn
67	P5	TDO	Misc.	O	Z	Z	JTAG Data Output	-
68	T3	VCLK	Video Output	O	L	L	Video Out Clock	-
69	V1	HSYNC	Video Output	O	L	L	H-Sync for RGB output	-
70	U2	VSYNC	Video Output	O	L	L	V-Sync/Composite-Sync output	-
71	N7	YOUT7	Video Output	I/O	L	L	Y OUT Signal	-
72	U3	YOUT6	Video Output	I/O	L	L	Y OUT Signal	-
73	V2	YOUT5	Video Output	I/O	L	L	Y OUT Signal	-
74	W2	YOUT4	Video Output	I/O	L	L	Y OUT Signal	-
75	V3	YOUT3	Video Output	I/O	L	L	Y OUT Signal	-
76	U4	YOUT2	Video Output	I/O	L	L	Y OUT Signal	-
77	W3	YOUT1	Video Output	I/O	L	L	Y OUT Signal	-
78	V4	YOUT0	Video Output	I/O	L	L	Y OUT Signal	-
79	P6	DVDD3	Power	-	-	-	VDD for I/O	-
80	J9	DGND3	GND	-	-	-	GND for I/O	-
81	W4	COUT7	Video Output	I/O	L	L	C OUT Signal	-
82	U5	COUT6	Video Output	I/O	L	L	C OUT Signal	-
83	V5	COUT5	Video Output	I/O	L	L	C OUT Signal	-
84	R6	COUT4	Video Output	I/O	L	L	C OUT Signal	-
85	W5	COUT3	Video Output	I/O	L	L	C OUT Signal	-
86	R7	COUT2	Video Output	I/O	L	L	C OUT Signal	-
87	U6	COUT1	Video Output	I/O	L	L	C OUT Signal	-
88	V6	COUT0	Video Output	I/O	L	L	C OUT Signal	-
89	W6	CVDD3	Power	-	-	-	VDD for core	-
90	J10	CGND3	GND	-	-	-	GND for core	-
91	N8	YI7	Video Input	I	I	I	Y IN Signal	Dn
92	R8	YI6	Video Input	I	I	I	Y IN Signal	Dn
93	U7	YI5	Video Input	I	I	I	Y IN Signal	Dn
94	V7	YI4	Video Input	I	I	I	Y IN Signal	Dn
95	W7	YI3	Video Input	I	I	I	Y IN Signal	Dn
96	N9	YI2	Video Input	I	I	I	Y IN Signal	Dn
97	P9	YI1	Video Input	I	I	I	Y IN Signal	Dn
98	U8	YI0	Video Input	I	I	I	Y IN Signal	Dn
99	J11	DGND4	GND	-	-	-	GND for I/O	-
100	V8	PCLK	Video Input	I	I	I	Pixel Clock	-
101	W8	DVDD4	Power	-	-	-	VDD for I/O	-
102	R9	HD	Video Input	I/O	I	I	H-Sync	-
103	M10	VD	Video Input	I/O	I	I	V-Sync	-
104	U9	FID	Video Input	I/O	I	I	Field Indicator	-
105	V9	CI7	Video Input	I	I	I	C IN Signal	Dn
106	W9	CI6	Video Input	I	I	I	C IN Signal	Dn
107	N10	CI5	Video Input	I	I	I	C IN Signal	Dn
108	R10	CI4	Video Input	I	I	I	C IN Signal	Dn
109	U10	CI3	Video Input	I	I	I	C IN Signal	Dn
110	V10	CI2	Video Input	I	I	I	C IN Signal	Dn
111	W10	CI1	Video Input	I	I	I	C IN Signal	Dn
112	P12	CI0	Video Input	I	I	I	C IN Signal	Dn
113	R11	PLLVDD1	Power	-	-	-	VDD for PLL	-
114	U11	PLLGND1	GND	-	-	-	GND for PLL	-
115	J12	CGND4	GND	-	-	-	GND for core	-
116	V11	M48XO	Misc.	O	-	-	48MHz X'tal output	-
117	W11	M48XI	Misc.	I/O	I	I	48MHz X'tal Input	-
118	U12	CVDD4	Power	-	-	-	VDD for core	-

PIN #	CSP BALL #	Pin Name	Group	Type	During reset		Pin Function	Pull up/ down
					Nrml	Ehost		
119	K8	DGND5	GND	-	-	-	GND for I/O	-
120	V12	SYSCLK	Misc.	I	I	I	External Clock Input	-
121	N11	DVDD5	Power	-	-	-	VDD for I/O	-
122	W12	BTSEL	Misc.	I	I	I	ARM Boot Selector	-
123	R12	FLSHSEL	Misc.	I	I	I	Flash ROM Bus Selecter	-
124	U13	EXTHOST	Misc.	I	I	I	EMIF/EHIF Selecter	-
125	V13	IREF	Video DAC	I	I	I	DAC Current Control	-
126	W13	DAOUT	Video DAC	O	O	O	DAC Composite Output	-
127	W14	VSSA	GND	-	-	-	analog GND for DAC	-
128	V14	VDDA	Power	-	-	-	analog VDD for DAC	-
129	U14	BIAS	Video DAC	I	I	I	DAC Bias with Compensation Capacitor	-
130	W15	VREFIN	Video DAC	I	I	I	DAC Reference Voltage	-
131	R13	VSS3V	GND	-	-	-	digital GND for DAC	-
132	V15	VDD3V	Power	-	-	-	digital VDD for DAC	-
133	U15	USB_VDD	Power	-	-	-	VDD for USB	-
134	W16	USB_DP	Communication I/F	I/O	I	I	USB Data Plus	-
135	W17	USB_DM	Communication I/F	I/O	I	I	USB Data Minus	-
136	V16	CVDD5	Power	-	-	-	VDD for core	-
137	K9	CGND5	GND	-	-	-	GND for core	-
138	R14	SDR_A14	SDRAM I/F	O	L	L	SDRAM Bank 0	-
139	U16	SDR_A13	SDRAM I/F	O	L	L	SDRAM Bank 1	-
140	V17	SDR_A12	SDRAM I/F	O	L	L	SDRAM Address(MSB)	-
141	W18	SDR_A11	SDRAM I/F	O	L	L	SDRAM Address	-
142	N12	SDR_A10	SDRAM I/F	O	L	L	SDRAM Address	-
143	U17	SDR_A9	SDRAM I/F	O	L	L	SDRAM Address	-
144	V18	SDR_A8	SDRAM I/F	O	L	L	SDRAM Address	-
145	N13	SDR_A7	SDRAM I/F	O	L	L	SDRAM Address	-
146	V19	SDR_A6	SDRAM I/F	O	L	L	SDRAM Address	-
147	U18	SDR_A5	SDRAM I/F	O	L	L	SDRAM Address	-
148	T17	SDR_A4	SDRAM I/F	O	L	L	SDRAM Address	-
149	U19	DVDDSD1	Power	-	-	-	VDD for SDRAM I/O	-
150	T19	DGNDSD1	GND	-	-	-	GND for SDRAM I/O	-
151	P14	SDR_A3	SDRAM I/F	O	L	L	SDRAM Address	-
152	T18	SDR_A2	SDRAM I/F	O	L	L	SDRAM Address	-
153	R17	SDR_A1	SDRAM I/F	O	L	L	SDRAM Address	-
154	P15	SDR_A0	SDRAM I/F	O	L	L	SDRAM Address(LSB)	-
155	R19	SDR_DQ31	SDRAM I/F	I/O	I	I	SDRAM Data (MSB)	Dn
156	M12	SDR_DQ30	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
157	R18	SDR_DQ29	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
158	P17	SDR_DQ28	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
159	N15	SDR_DQ27	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
160	P19	SDR_DQ26	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
161	M13	SDR_DQ25	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
162	P18	SDR_DQ24	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
163	N17	CVDD6	Power	-	-	-	VDD for core	-
164	K10	CGND6	GND	-	-	-	GND for core	-
165	M15	SDR_DQ23	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
166	N19	SDR_DQ22	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
167	M14	SDR_DQ21	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
168	N18	SDR_DQ20	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
169	M17	SDR_DQ19	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
170	L13	SDR_DQ18	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
171	M19	SDR_DQ17	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
172	M18	SDR_DQ16	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
173	L15	SDR_DQ15	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
174	L17	SDR_DQ14	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
175	L18	SDR_DQ13	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
176	L19	SDR_DQ12	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
177	K18	DVDDSD2	Power	-	-	-	VDD for SDRAM I/O	-
178	K19	DGNDSD2	GND	-	-	-	GND for SDRAM I/O	-
179	K17	SDR_DQ11	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
180	K15	SDR_DQ10	SDRAM I/F	I/O	I	I	SDRAM Data	Dn

PIN #	CSP BALL #	Pin Name	Group	Type	During reset Nrml	During reset Ehost	Pin Function	Pull up/ down
181	K14	SDR_DQ9	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
182	J18	SDR_DQ8	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
183	J19	SDR_DQ7	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
184	J17	SDR_DQ6	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
185	K13	SDR_DQ5	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
186	J15	SDR_DQ4	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
187	H18	SDR_DQ3	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
188	H19	SDR_DQ2	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
189	H17	SDR_DQ1	SDRAM I/F	I/O	I	I	SDRAM Data	Dn
190	G19	SDR_DQ0	SDRAM I/F	I/O	I	I	SDRAM Data (LSB)	Dn
191	J13	CVDD7	Power	-	-	-	VDD for core	-
192	K11	CGND7	GND	-	-	-	GND for core	-
193	G18	SDR_RAS	SDRAM I/F	O	H	H	SDRAM RAS Strobe	-
194	H15	SDR_CAS	SDRAM I/F	O	H	H	SDRAM CAS Strobe	-
195	G17	SDR_WE	SDRAM I/F	O	H	H	SDRAM WE	-
196	F19	SDR_CLK	SDRAM I/F	I/O	L	L	SDRAM I/F	-
197	H14	SDR_CS	SDRAM I/F	O	H	H	SDRAM CS	-
198	F18	SDR_CKE	SDRAM I/F	O	H	H	SDRAM CKE	-
199	G15	SDR_DQMHH	SDRAM I/F	O	H	H	DQMH for SDR_DQ[31:24]	-
200	F17	SDR_DQMHL	SDRAM I/F	O	H	H	DQMH for SDR_DQ[23:16]	-
201	E19	SDR_DQMLH	SDRAM I/F	O	H	H	DQMH for SDR_DQ[15:8]	-
202	E18	SDR_DQMLL	SDRAM I/F	O	H	H	DQMH for SDR_DQ[7:0]	-
203	H13	CFOE	External Memory I/F	O	H	H	CFC Output Enable	-
204	D19	CFE1	External Memory I/F	O	H	H	CFC Enable 1	-
205	E17	CFE2	External Memory I/F	O	H	H	CFC Enable 2	-
206	D18	CFWAIT	External Memory I/F	I	I	I	CFC Wait Signal Input	Up
207	F15	CFRDY	External Memory I/F	I	I	I	CFC Ready	Up
208	C19	IOIS16	External Memory I/F	I	I	I	CFC I/O Select	Up
209	C18	CFIORD	External Memory I/F	O	H	H	CFC I/O Read Strobe	-
210	D17	CFIOWR	External Memory I/F	O	H	H	CFC I/O Write Strobe	-
211	B19	DVDD6	Power	-	-	-	VDD for I/O	-
212	K12	DGND6	GND	-	-	-	GND for I/O	-
213	B18	SD_DATA1	SD/MS Card I/F	I/O	I	I	SD Data 1	-
214	C17	SD_DATA0	SD/MS Card I/F	I/O	I	I	SD Data 0	-
215	A18	SD_CLK	SD/MS Card I/F	O	L	L	SD Clock	-
216	B17	SD_CMD	SD/MS Card I/F	I/O	I	I	SD Command	-
217	G13	SD_DATA3	SD/MS Card I/F	I/O	I	I	SD Data 3	-
218	C16	SD_DATA2	SD/MS Card I/F	I/O	I	I	SD Data 2	-
219	A17	CVDD8	Power	-	-	-	VDD for core	-
220	L9	CGND8	GND	-	-	-	GND for core	-
221	B16	EM_WE	External Memory I/F	I/O	H	I	External Memory Write Enable	-
222	A16	EM_OE	External Memory I/F	I/O	H	I	External Memory Output Enable	-
223	F14	EM_WAIT	External Memory I/F	I/O	I	I	External Memory Wait	-
224	C15	EM_BEL	External Memory I/F	O	H	H	External Memory Byte Enable Low	-
225	E14	EM_BEH	External Memory I/F	O	H	H	External Memory Byte Enable High	-
226	B15	EM_CE0	External Memory I/F	I/O	H	I	External Memory Chip Enable #1	-
227	A15	EM_CE1	External Memory I/F	O	H	H	External Memory Chip Enable #2	-
228	C14	EM_RDWR	External Memory I/F	O	H	H	External Memory Read/Write	-
229	E13	DVDD7	Power	-	-	-	VDD for I/O	-
230	B14	MXO	Misc.	O	-	-	X'tal Output	-
231	A14	MXI	Misc.	I/O	-	-	X'tal Input	-
232	L10	DGND7	GND	-	-	-	GND for I/O	-
233	B13	PLLVDD2	Power	-	-	-	VDD for PLL	-
234	C13	PLLGND2	GND	-	-	-	GND for PLL	-
235	G12	TEST4	Misc.	I	I	I	Test Input4 (INITZ)	Up
236	A13	FLSH_CE	External Memory I/F	O	H	H	Flash ROM Chip Enable	-
237	E12	BUSWIDTH	External Memory I/F	I	I	I	8/16 bit select for FLSH_CEH	-
238	B12	CVDD9	Power	-	-	-	VDD for core	-
239	L11	CGND9	GND	-	-	-	GND for core	-
240	C12	ARM_D15	External Memory I/F	I/O	I	I	External Memory Data (MSB)	-
241	G11	ARM_D14	External Memory I/F	I/O	I	I	External Memory Data	-
242	A12	ARM_D13	External Memory I/F	I/O	I	I	External Memory Data	-

PIN #	CSP BALL #	Pin Name	Group	Type	During reset		Pin Function	Pull up/ down
					Nrml	Ehost		
243	F12	ARM_D12	External Memory I/F	I/O	I	I	External Memory Data	-
244	B11	ARM_D11	External Memory I/F	I/O	I	I	External Memory Data	-
245	C11	ARM_D10	External Memory I/F	I/O	I	I	External Memory Data	-
246	A11	ARM_D9	External Memory I/F	I/O	I	I	External Memory Data	-
247	E11	ARM_D8	External Memory I/F	I/O	I	I	External Memory Data	-
248	A10	ARM_D7	External Memory I/F	I/O	I	I	External Memory Data	-
249	B10	DVDD8	Power	-	-	-	VDD for I/O	-
250	L12	DGND8	GND	-	-	-	GND for I/O	-
251	C10	ARM_D6	External Memory I/F	I/O	I	I	External Memory Data	-
252	G10	ARM_D5	External Memory I/F	I/O	I	I	External Memory Data	-
253	E10	ARM_D4	External Memory I/F	I/O	I	I	External Memory Data	-
254	A9	ARM_D3	External Memory I/F	I/O	I	I	External Memory Data	-
255	B9	ARM_D2	External Memory I/F	I/O	I	I	External Memory Data	-
256	C9	ARM_D1	External Memory I/F	I/O	I	I	External Memory Data	-
257	F10	ARM_D0	External Memory I/F	I/O	I	I	External Memory Data (LSB)	-
258	A8	CVDD10	Power	-	-	-	VDD for core	-
259	M8	CGND10	GND	-	-	-	GND for core	-
260	E9	ARM_A22	External Memory I/F	I/O	L	I	External Memory Address (MSB)	-
261	B8	ARM_A21	External Memory I/F	I/O	L	I	External Memory Address	-
262	C8	ARM_A20	External Memory I/F	I/O	L	I	External Memory Address	-
263	F9	ARM_A19	External Memory I/F	I/O	L	I	External Memory Address	-
264	A7	ARM_A18	External Memory I/F	I/O	L	I	External Memory Address	-
265	G9	ARM_A17	External Memory I/F	I/O	L	I	External Memory Address	-
266	B7	ARM_A16	External Memory I/F	I/O	L	I	External Memory Address	-
267	C7	ARM_A15	External Memory I/F	I/O	L	I	External Memory Address	-
268	E8	ARM_A14	External Memory I/F	I/O	L	I	External Memory Address	-
269	A6	DVDD9	Power	-	-	-	VDD for I/O	-
270	M9	DGND9	GND	-	-	-	GND for I/O	-
271	H9	ARM_A13	External Memory I/F	I/O	L	I	External Memory Address	-
272	B6	ARM_A12	External Memory I/F	I/O	L	I	External Memory Address	-
273	C6	ARM_A11	External Memory I/F	I/O	L	I	External Memory Address	-
274	A5	ARM_A10	External Memory I/F	I/O	L	I	External Memory Address	-
275	E7	ARM_A9	External Memory I/F	I/O	L	I	External Memory Address	-
276	B5	ARM_A8	External Memory I/F	I/O	L	I	External Memory Address	-
277	G8	ARM_A7	External Memory I/F	I/O	L	I	External Memory Address	-
278	C5	ARM_A6	External Memory I/F	I/O	L	I	External Memory Address	-
279	A4	CVDD11	Power	-	-	-	VDD for core	-
280	M11	CGND11	GND	-	-	-	GND for core	-
281	E6	ARM_A5	External Memory I/F	I/O	L	I	External Memory Address	-
282	B4	ARM_A4	External Memory I/F	I/O	L	I	External Memory Address	-
283	A3	ARM_A3	External Memory I/F	I/O	L	I	External Memory Address	-
284	C4	ARM_A2	External Memory I/F	I/O	L	I	External Memory Address	-
285	G7	ARM_A1	External Memory I/F	I/O	L	I	External Memory Address	-
286	B3	ARM_A0	External Memory I/F	I/O	L	I	External Memory Address (LSB)	-
287	A2	EMU0	Misc.	I/O	I	I	Emulator Interrupt0	-
288	C3	EMU1	Misc.	I/O	I	I	Emulator Interrupt1	-

note *1) Following pins are connected to common GND.
H8, H10, H11, H12, J9, J10, J11, J12, K8, K9, K10, K11, K12, L9, L10, L11, L12, M8, M9, M11

note *2)
Dn Pulled down during reset.

3. Parameter Information

3.1 Absolute maximum ratings

PARAMTER	MIN	MAX	UNIT
Supply voltage range for core, CVDD	-0.5	1.836	V
Supply voltage range for I/O, DVDD	-0.5	4.0	V
Supply voltage range for PLL, PLLVDD	-0.5	1.836	V
Supply voltage for D/A, VDD3V	-0.5	4.0	V
Supply voltage for D/A, VDDA	-0.5	4.0	V
Supply voltage for USB, USBVDD	-0.5	4.0	V
Input voltage range for input, VI	-0.5	DVDD+0.5 CVDD+0.5 USBVDD+0.5	V
Output voltage range for output, VO	-0.5	DVDD+0.5 CVDD+0.5 USBVDD+0.5	V
Clamp current for an input or output	-20	+20	mA
Storage temperature range, Tstg	-65	150	℃

1. Functional operation of the device is not implied at any conditions beyond those listed under recommended operating conditions.
2. Additional absolute maximum ratings for specific I/O functions are listed throughout this document.
3. Conditions in the field that exceed any of the absolute maximum ratings for a particular macro can cause permanent damage to the device.
4. Device reliability may be affected by exposure to absolute-maximum-rated conditions for extended periods.
5. Clamp current flows from an input or output pad to a supply rail through a clamp circuit or an intrinsic diode. Positive current results from an applied input or output voltage that is more than 0.5 V higher (more positive) than the supply voltage, CVDD for single-supply macros or DVDD for dual-supply macros. Negative current results from an applied voltage that is more than 0.5 V less (more negative) than the DVSS or CVSS voltage.

3.2 Recommended operating conditions (DVDD > CVDD)

	PARAMETER	MIN	NOM	MAX	UNIT
CVDD	Supply voltage, Core	1.40	1.50	1.60	V
PLLVDD	Supply voltage, PLL	1.40	1.50	1.60	V
DVDD	Supply voltage, I/O	**2.70**		3.60	V
DGND, CGND,PLLGND	Supply voltage, GROUND		0		V
VI(OSC)	Input voltage, OSC (*1)	0		CVDD	V
VI(Other)	Input voltage, Other pins (*2)	0		DVDD	V
VO(OSC)	Output voltage, OSC (*1)	0		CVDD	V
VO(Other)	Output voltage, Other pins (*2)	0		DVDD	V
VIH(OSC)	High-level dc input voltage, OSC (*1)	0.7CVDD			V
VIH(Other)	High-level dc input voltage, Other pins (*2)	0.7DVDD			V
VIL(OSC)	Low-level dc input voltage, OSC (*1)	0		0.3CVDD	V
VIL(Other)	Low-level dc input voltage, Other pins (*2)	0		0.3DVDD	V
Vhys	Hysteresis voltage, Pins : (*3)		0.13DVDD		V
Ii	Input current, VI=0 to MAX	-1		+1	μA
Tc	Operating case temperature range	**-20**		85	℃

*1) OSC : MXI/O, M48XI/O
*2) Other pins : Other than *1) pins
*3) WAKEUP, GIO1-33,RSN,PWDWN,TMS,TRST,TEST0-4,SCANEN

3.3 Recommended operating conditions for USB (*1)

	PARAMETER	MIN	NOM	MAX	UNIT
USB_VDD	Supply voltage	**3.00**	3.30	**3.60**	V
VI (Single-ended)	Input voltage, (*2)	0		USB_VDD	V
VI (Differential)	Input voltage, (*3)	0.80		2.50	V
VIH (Single-ended)	High-level dc input voltage, (*2)	2.00			V
VIL (Single-ended)	Low-level dc input voltage, (*2)			0.8	V
VID	Differential input voltage, (*3)	200			mV
VO	Output voltage	0		USB_VDD	V
RS	Series resistance (5% tolerance)		27		Ω

*1) Compatible with USB(Universal Serial Bus Interface) specifications, Rev1.1, Sep23, 1998
*2) Single-ended input buffers with hysteresis. Compatible with TTL signal levels
*3) Differential input buffers

3.4 Recommended operating conditions for D/A Converter

	PARAMETER	MIN	NOM	MAX	UNIT
VDD3V	Supply voltage, (digital part)	**3.0**	3.3	**3.6**	V
VDDA	Supply voltage, (analog part)	**3.0**	3.3	**3.6**	V
VSS3V	Supply voltage, (digital part)		0		V
VSSA	Supply voltage, (analog part)		0		V
VO (*1)	Output voltage	VSSA		Vref	V
Vref	Reference voltage range	1		1.5	V
Riref	Reference register		1.6		kΩ
Raout	Output resgister		200		Ω
Cbias	Compensation capacitor BIAS to VSSA		0.1		μF

*1) VO range is kept while the ratio (Riref:Raout) have 8:1 relationship.

3.5 Parameter measurement information

Where: Iol =2.0mA (All outputs)
Ioh = 2.0mA (All outputs)
Vth = VDDx0.5V
Ct = 45pF typical load circuit capacitance

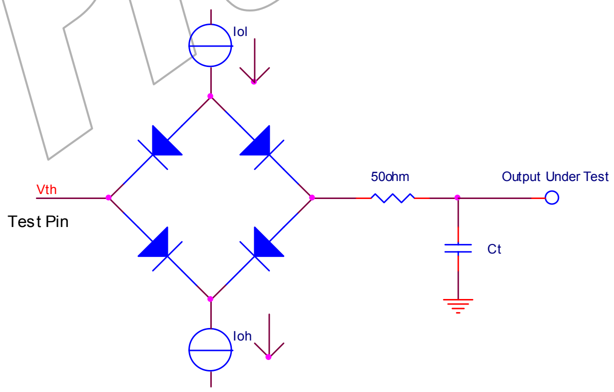


Figure . Test Load circuit

3.6 Electrical characteristics over recommended operating conditions

	PARAMETER	CONDITIONS	MIN	MAX	UNIT
VOH(OSC)	High-level output voltage, OSC(*1)	IOH=Rated	0.8CVDD		V
VOH(USB)	High-level output voltage, USB(*2)	IOH=-18.3mA	USB_VDD-0.5		V
VOH(other)	High-level output voltage, Other(*3)	IOH=4mA	0.8DVDD		V
VOL(OSC)	Low-level output voltage, OSC(*1)	IOL=Rated		0.22CVDD	V
VOL(USB)	Low-level output voltage, USB(*2)	IOL=18.3mA		0.28	V
VOL(other)	Low-level output voltage, Other(*3)	IOL=4mA		0.22DVDD	V
IIH	Input Leakage	VI = VI max	-1	+1	μA
IIL	Input Leakage	VI = VI max	-1	+1	μA
Ioz	Output High-impedance Leakage		-20	+20	μA
Ci	Input capacitance			10	pF
Co	Output capacitance			10	pF

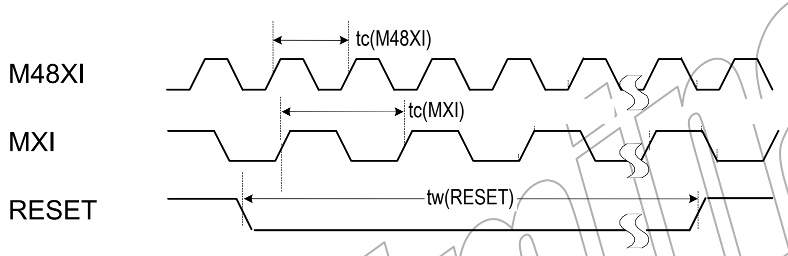
*1) OSC : MXI/O, M48XI/O
*2) USB : USB_DP/DM
*3) Other pins : Other than *1) and *2) pins

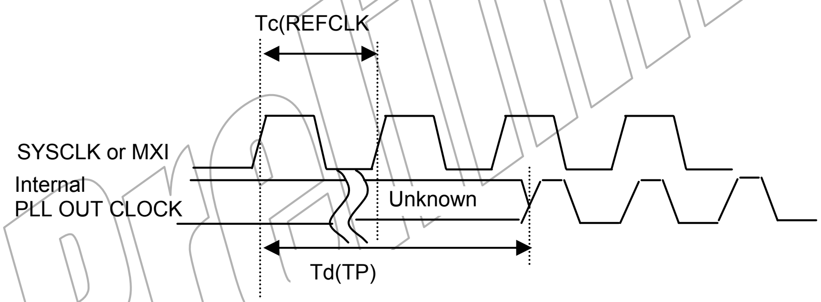
4. Timing Diagrams

4.1 OSC Interface and Clock Timing

	PARAMETER	MIN	NOM	MAX	UNIT
tc(M48XI)	Cycle time, M48XI (*1)			**48**	MHz
tc(MXI)	Cycle time, MXI (*2)			**27**	MHz
tc(SYSCLK)	Cycle time, external SYSCLK			**40**	MHz
tc(ARM)	Cycle time, internal ARM clock (*4)			**88**	MHz
tc(SDR)	Cycle time, internal SDRAM clock (*4,*6)			**102**	MHz
tc(DSP)	Cycle time, internal DSP clock (*4)			**102**	MHz
tc(iMX)	Cycle time, internal iMX clock (*4)			**180**	MHz
tw(RESET)	Pulse duration, RESET low (*3)	**10cycle(tc(MXI))**			ns
Td(TP)	PLL lock-in period from clock in	**500 (*5)**			us

*1) For USB and Pseudo-I2C.
*2) For Video output.
*3) RESET need 10cycle (tc(ARM)) after Cycle stable.
*4) Each cycle time of internal module is controlled by the register on CLKC module. (Please refer “DM270 register manual”.)
*5) Under condition the reference clock to PLL = 27MHz
*6) The maximum performance of SDRAM shows the characterization data on EVM.





4.2 Audio Interface Timing

timing requirements for McBSP [H=0.5t c(DSP)]†

PARAMETER			MIN	MAX	UNIT
tc(BCKRX)	Cycle time, BCLKR/X	BCLKR/X ext	4H*		ns
th(BCKRL-BFRH)	Hold time, external BFSR high after BCLKR low	BCLKR int	0		ns
		BCLKR ext	4		
th(BCKRL-BDRV)	Hold time, BDR valid after BCLKR low	BCLKR int	0		ns
		BCLKR ext	6		
th(BCKXL-BFXH)	Hold time, external BFSX high after BCLKX low	BCLKX int	0		ns
		BCLKX ext	10		
tsu(BFRH-BCKRL)	Setup time, external BFSR high before BCLKR low	BCLKR int	20		ns
		BCLKR ext	5		
tsu(BDRV-BCKRL)	Setup time, BDR valid before BCLKR low	BCLKR int	20		ns
		BCLKR ext	4		
tsu(BFXH-BCKXL)	Setup time, external BFSX high before BCLKX low	BCLKX int	20		ns
		BCLKX ext	6		

†Polarity bits CLKRP = CLKXP = FSRP = FSXP = 0. If the polarity of any of the signals is inverted, then the timing references of that signal are also inverted.

*H is a half of DSP CLOCK.

switching characteristics for McBSP [H=0.5t c(DSP)]†

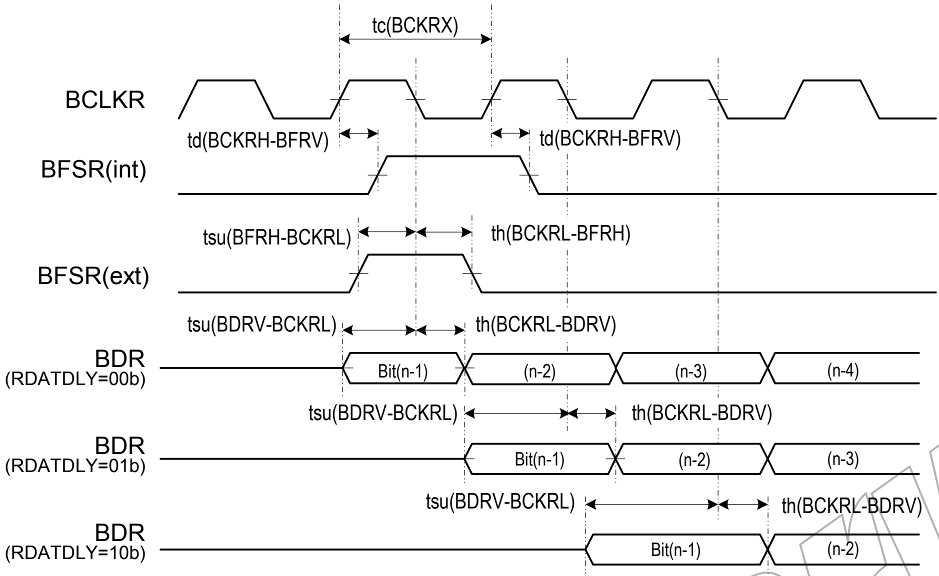
PARAMETER				MIN	MAX	UNIT
tc(CLKS)	Cycle time			4H*		ns
tc(BCKRX)	Cycle time, BCLKR/X		BCLKR/X int	4H*		ns
td (BCKRH-BFRV)	Delay time, BCLKR high to internal BFSR valid		BCLKR int	-4	6	ns
td(BCKXH-BFXV)	Delay time, BCLKX high to internal BFSX valid		BCLKX int	-2	11	ns
			BCLKX ext	4	23	
tdis(BCKXH-BDXHZ)	Disable time, BCLKX high to BDX high impedance following last data bit		BCLKX int	-15	1‡	ns
			BCLKX ext	-2	23‡	
td(BCKXH-BDXV)	Delay time, BCLKX high to BDX valid. This applies to all bits except the first bit transmitted.		BCLKXi nt	-10	8	ns
			BCLKX ext	2	24	
	Delay time, BCLKX high to BDX valid. Only applies to first bit transmitted when in Data Delay 1 or 2(XDATDLY=01b or 10b) modes	DXENA=0	BCLKX int		10	ns
			BCLKX ext		24	
te(BCKXH-BDX)	Enable time, BCLKX high to BDX driven. Only applies to first bit transmitted when in Data Delay 1 or 2 (XDATDLY = 01b or 10b) modes.	DXENA=0	BCLKX int	-2‡		ns
			BCLKX ext	2‡		
td(BFXH-BDXV)	Delay time, BFSX high to BDX valid. Only applies to first bit transmitted when in Data Delay 0 (XDATDLY = 00b) mode.	DXENA=0	BFSX int		5	ns
			BFSX ext		20	
te(BFXH-BDX)	Enable time, BFSX high to BDX driven. Only applies to first bit transmitted when in Data Delay 0 (XDATDLY = 00b) modes.	DXENA=0	BFSX int	-3		ns
			BFSX ext	2		
td(CLKS-BCLKRX)	Delay time,CLKS high to internal BCLKR/X	BCLKR/X	BCLKR/X int	3	26	ns

†Polarity bits CLKRP = CLKXP = FSRP = FSXP = 0. If the polarity of any of the signals is inverted, then the timing references of that signal are also inverted.

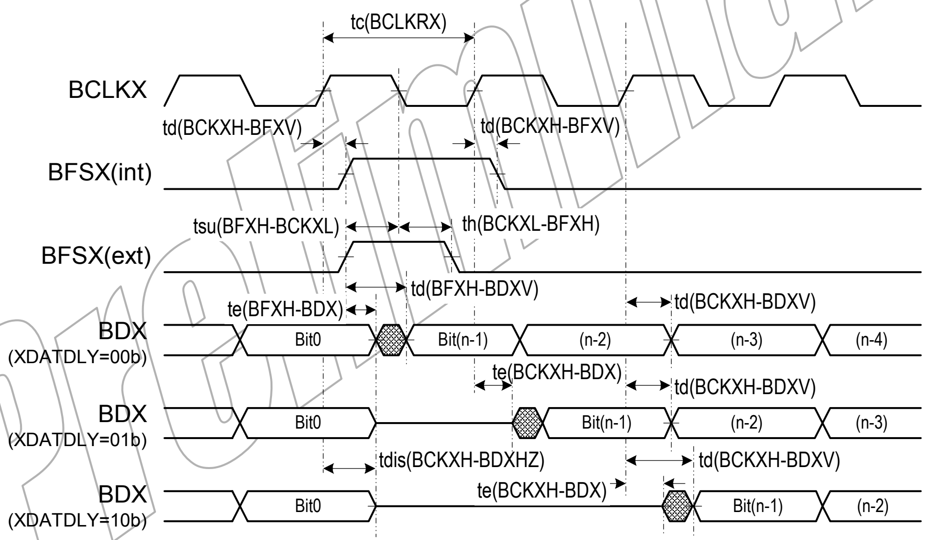
‡ Value assured by design but not tested.

* H is a half of DSP CLOCK

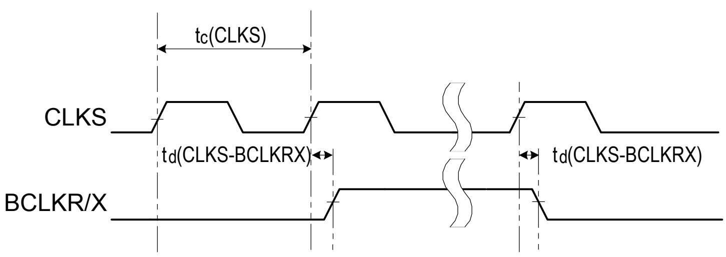
McBSP receive timings



McBSP transmit timings



CLKS Timings



4.3 Video Interface Timing

PCLK is used as clock source of following parameters

PARAMETER		MIN	NOM	MAX	UNIT
tCLK	PCLK			40	Mhz
tCD	VCLK	2.5		15.5	ns
tS	CI setup time	4.5			ns
tS	COUT setup time	4.5			ns
tS	YI setup time	4.5			ns
tS	YOUT setup time	4.5			ns
tS	HD setup time	4.5			ns
tS	VD setup time	4.5			ns
tS	GIO26(C_WEN) setup time	4.5			ns
tH	CI hold time	3.5			ns
tH	COUT hold time	3.5			ns
tH	YI hold time	3.5			ns
tH	YOUT hold time	3.5			ns
tH	HD hold time	3.5			ns
tH	VD hold time	3.5			ns
tH	GIO26(C_WEN) hold time	3.5			ns
tPD	COUT delay time	0.5		15.5	ns
tPD	YOUT delay time	0.5		15.5	ns
tPD	HD delay time	0.5		15.5	ns
tPD	VD delay time	0.5		15.5	ns
tPD	HSYNC delay time	0.5		15.5	ns
tPD	VSYNC delay time	0.5		15.5	ns

SYSCLK is used as clock source of following parameters.

PARAMETER		MIN	NOM	MAX	UNIT
tCLK	PCLK			40	Mhz
tCD	VCLK	2.5		15.5	ns
tS	CI setup time	4.5			ns
tS	COUT setup time	4.5			ns
tS	YI setup time	4.5			ns
tS	YOUT setup time	4.5			ns
tS	HD setup time	4.5			ns
tS	VD setup time	4.5			ns
tS	GIO26(C_WEN) setup time	4.5			ns
tH	CI hold time	3.5			ns
tH	COUT hold time	3.5			ns
tH	YI hold time	3.5			ns
tH	YOUT hold time	3.5			ns
tH	HD hold time	3.5			ns
tH	VD hold time	3.5			ns
tH	GIO26(C_WEN) hold time	3.5			ns
tPD	COUT delay time	0.5		15.5	ns
tPD	YOUT delay time	0.5		15.5	ns
tPD	HD delay time	0.5		15.5	ns
tPD	VD delay time	0.5		15.5	ns
tPD	HSYNC delay time	0.5		15.5	ns
tPD	VSYNC delay time	0.5		15.5	ns

Note) tCLK : Clock cycle time
tCD : Output clock delay time
tS : Data setup time
tH : Data hold time
tPD : Output delay time

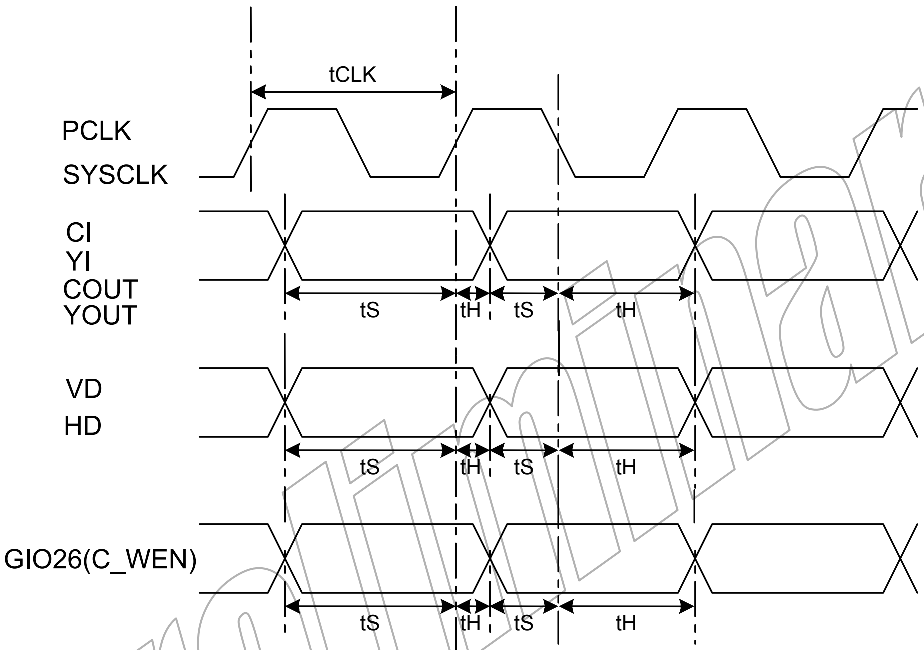
Note) these values of parameters from (STA +2ns or –2ns +-alpha)

Note) TDL name have a prefix "TDLAC02".

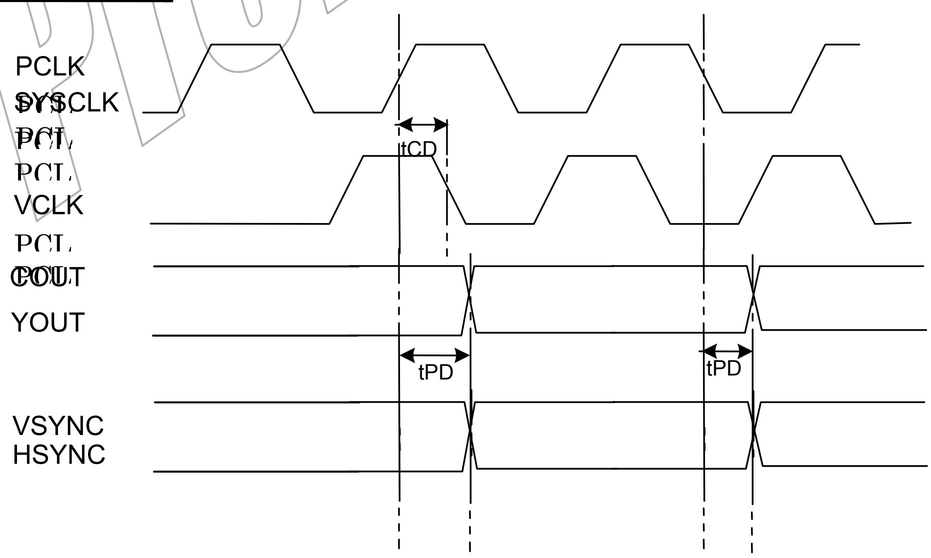
Video I/F Pin function Table (LCD Output)

Pin Name	*Standard*	*EPSON*	*CASIO*
YOUT7	Data7	Data5	Data5
YOUT6	Data6	Data4	Data4
YOUT5	Data5	Data3	Data3
YOUT4	Data4	Data2	Data2
YOUT3	Data3	Data1	Data1
YOUT2	Data2	Data0	Data0
YOUT1	Data1	GCP	STBYB
YOUT0	Data0	LP	RIT
COUT7	LCD_AC	RES	CP
COUT6	LCD_OE	XINH	STH
COUT5	BRIGHT	YSCL	STB
COUT4	COMP_SYNC	YSCLD	GPCK
COUT3	PWM_OUT	FRYS	GRES
COUT2	reserved	FRYP	POL
COUT1	reserved	FRX	reserved
COUT0	reserved	DY	GSRT

Input Timing



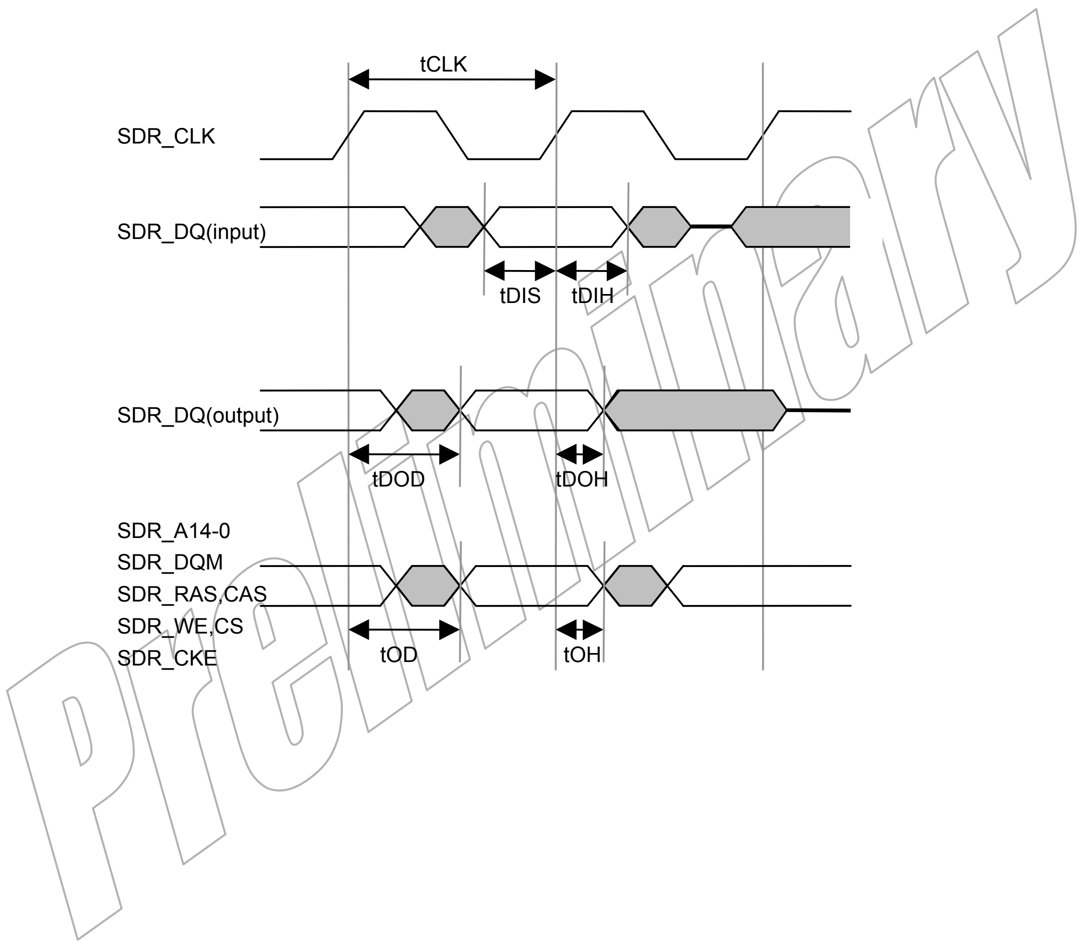
Output Timing



4.4 SDRAM Interface Timing

	PARAMETER	MIN	NOM	MAX	UNIT
tCLK	Cycle time, SDR_CLK (*1)	9.8		-	ns
tDIS	Input Data setup time (*2)	2.0			ns
tDIH	Input Data hold time (*2)	0.0			ns
tDOD	Output Data delay time (*2)			4.5	ns
tDOH	Output Data hold time (*2)	1.5			ns
tOD	Output Signal delay time (*2)			4.5	ns
tOH	Output Signal hold time (*2)	1.5			ns

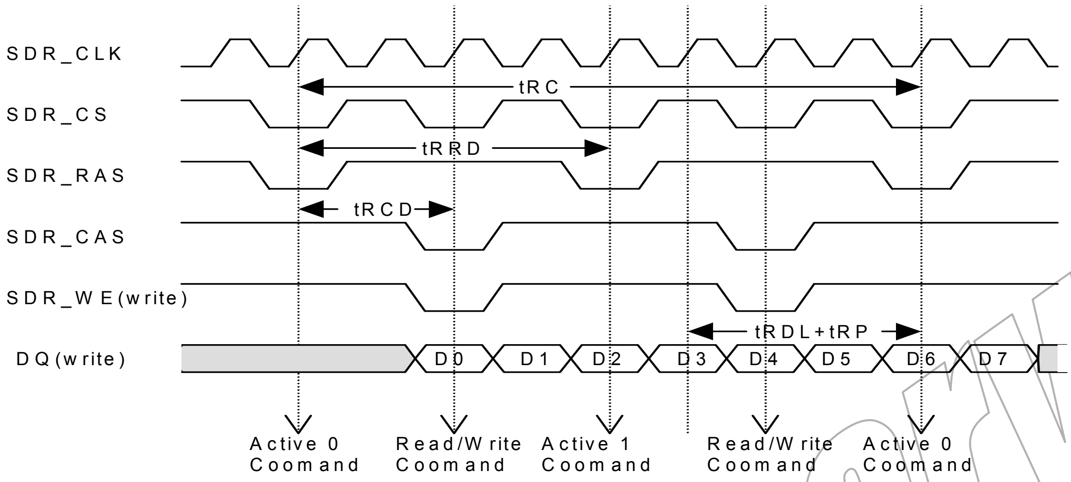
*1) This data shows the characterization data on EVM.
*2) These data show the design simulation data (not test load circuit).



Burst Read/Write with Auto-Precharge (BL = 4)

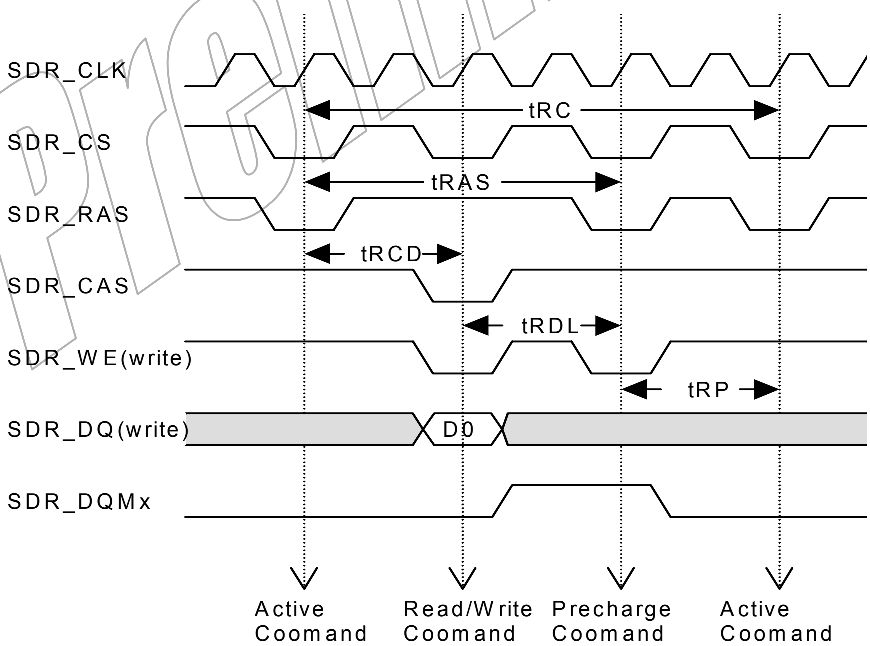
	PARAMTER	CYCLE	UNIT
tRC	Row cycle time	8(9)	CLK
tRCD	Row to Column delay	2	CLK
tRRD	Act to Act delay time	4	CLK
tRDL+tRP	Last data in to precharge + Row precharge time	3(4)	CLK

(*): TRDL = 1(MODESET register)



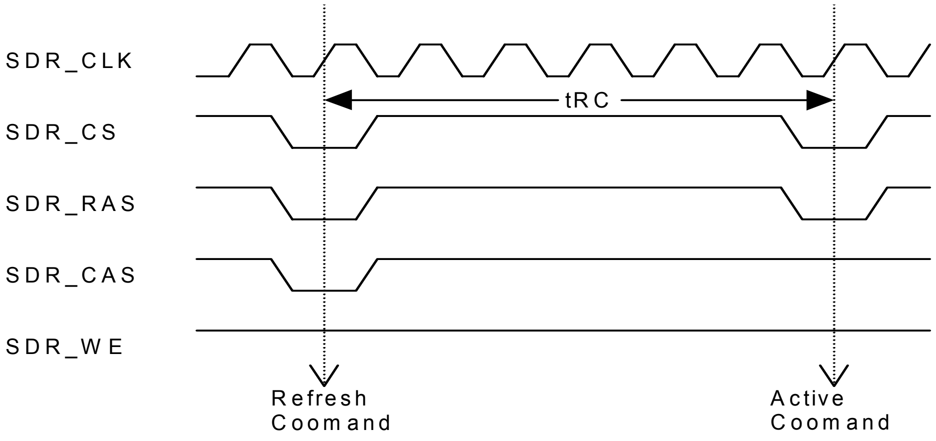
Burst Interrupt by Precharge (BL = 4)

	PARAMTER	CYCLE	UNIT
tRC	Row cycle time	7	CLK
tRCD	Row to Column delay	2	CLK
tRAS	Row Active time	5	CLK
tRDL	Last data in to precharge	3	CLK
tRP	Row precharge time	2	CLK



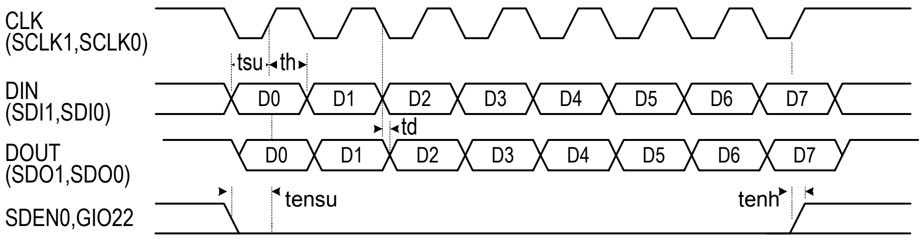
Auto Refresh

	PARAMTER	CYCLE	UNIT
tRC	Row cycle time	7	CLK



4.5 Serial Interface Timing

PARAMTER		MIN	NOM	MAX	UNIT
Tsu	Setup time	3			ns
th	Hold time	4			ns
td	Delay time			18	ns
tensu	Enable setup time	3			ns
tenh	Enable hold time	4			ns

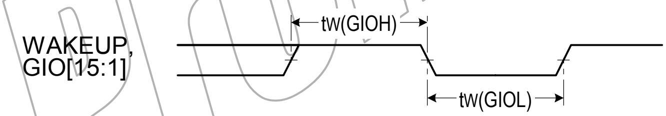


4.6 Wakeup, General I/O (Interrupt Input) Timing

PARAMTER		MIN	NOM	MAX	UNIT
tw(GIOH)	WAKEUP, GIO[15:1] high pulse width	2Tcyc*			ns
tw(GIOL)	WAKEUP, GIO[15:1] low pulse width	2Tcyc*			ns

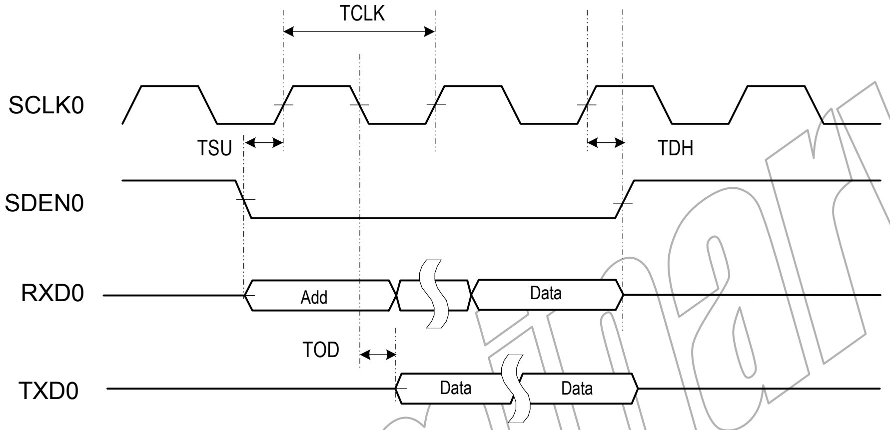
* Tcyc means one cycle time of ARM clock.

GIO external interrupt timings



4.7 AIM Serial Interface Timing

	PARAMTER	MIN	NOM	MAX	UNIT
TCLK	Cycle time, (SCLK0)	100			ns
TSU	Input setup, (SDEN0, RXD0)	1			ns
TDH	Input hold, (SDEN0, RXD0)	5			ns
TOD	Output signal delay time, (TXD0)			18	ns

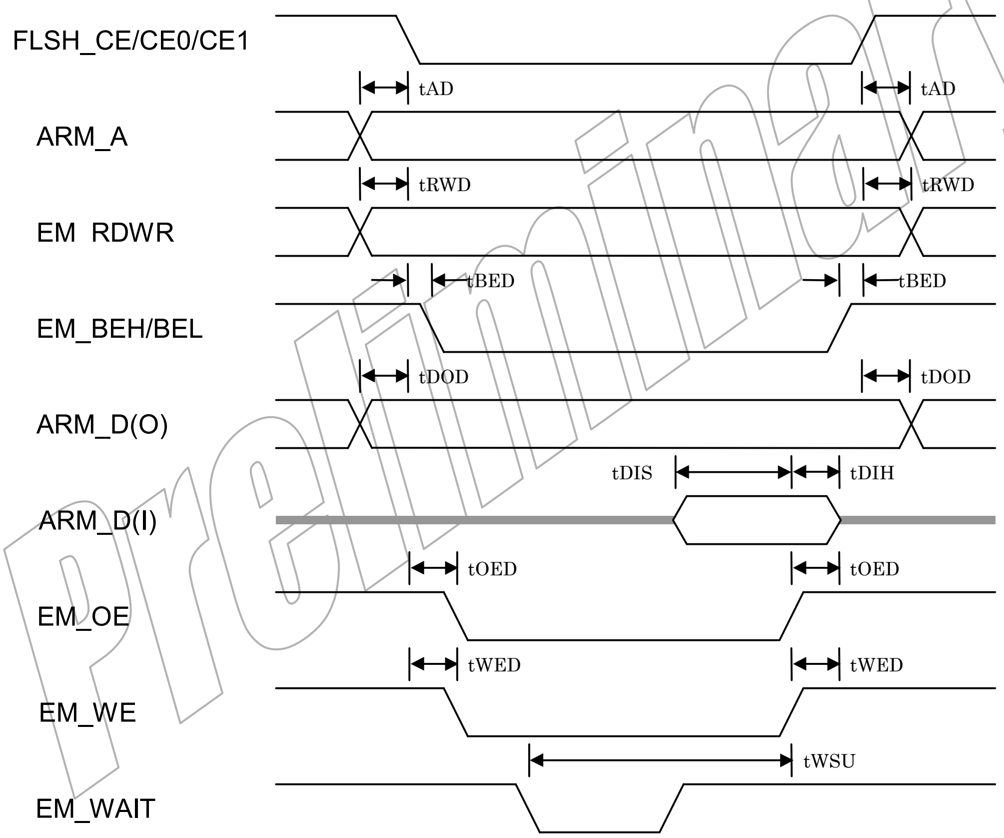


4.8 External Memory Interface Timing

4.8.1 FLASH/SRAM Interface

	PARAMETER	MIN	NOM	MAX	UNIT
tAD	Address to CE delay time	NxTc - 3		NxTc + 7.5	ns
tRWD	RW to CE delay time	NxTc - 3		NxTc + 7	ns
tBED	CE to BE delay time	NxTc - 3		NxTc + 7	ns
tDOD	outut data to CE delay time	NxTc - 9.5		NxTc + 4	ns
tDIS	input data setup time to OE de-asserted	8			ns
tDIH	input data hold time from OE de-asserted	0			ns
tOED	CE to OE delay time	NxTc - 3		NxTc + 7	ns
tWED	CE to WE hold time	NxTc - 3		NxTc + 5	ns
tWSU	WAIT setup time to OE/WE de-aserted	2xTc+6			ns

FLASH&SRAM AC timing

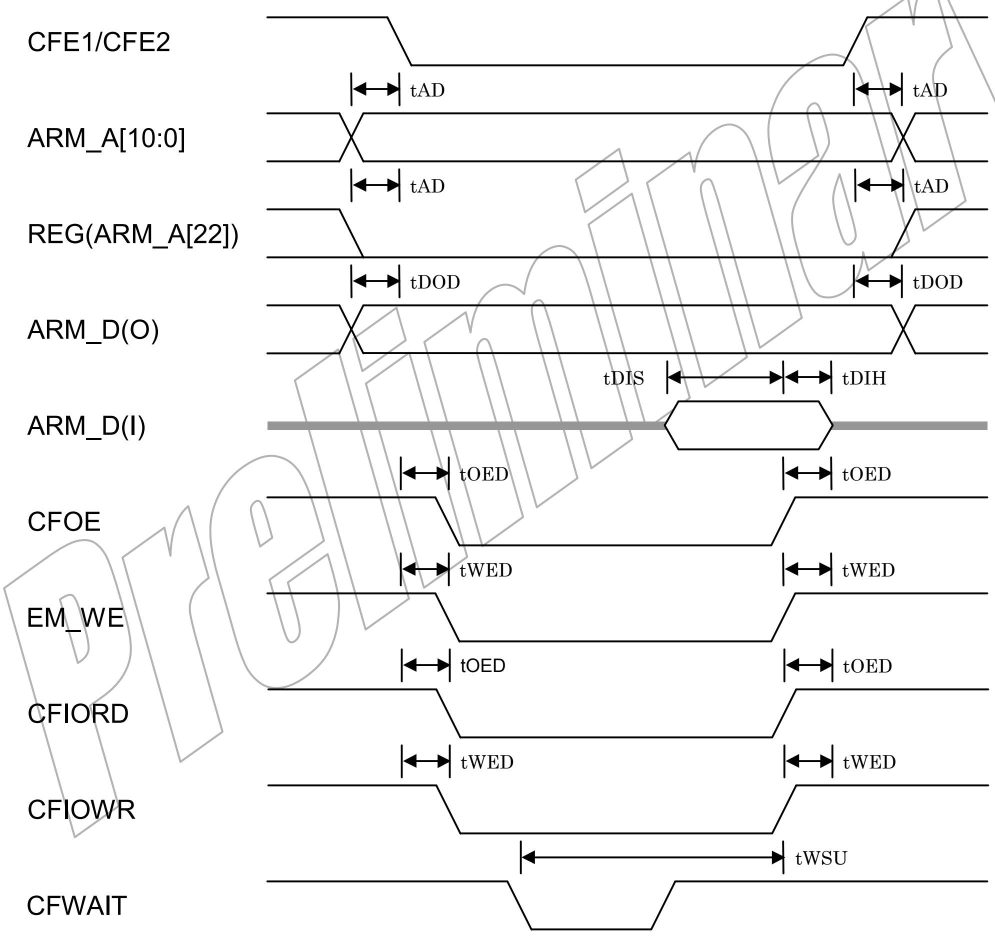


4.8.2 COMPACT FLASH Interface

	PARAMETER	MIN	NOM	MAX	UNIT
tAD	Address to CE delay time	NxTc - 3		NxTc +7.5	ns
tDOD	outut data to CE delay time	NxTc - 9.5		NxTc + 4	ns
tDIS	input data setup time to OE de-asserted	8			ns
tDIH	input data hold time from OE de-asserted	0			ns
tOED	CE to OE delay time	NxTc - 3		NxTc + 7	ns
tWED	CE to WE hold time	NxTc - 3		NxTc + 5	ns
tWSU	WAIT setup time to OE/WE de-aserted	2xTc+6			ns

Tc: ARM clock cycle
N : Programmed number of ARM clocks by specific parameter

CompactFlash AC timing

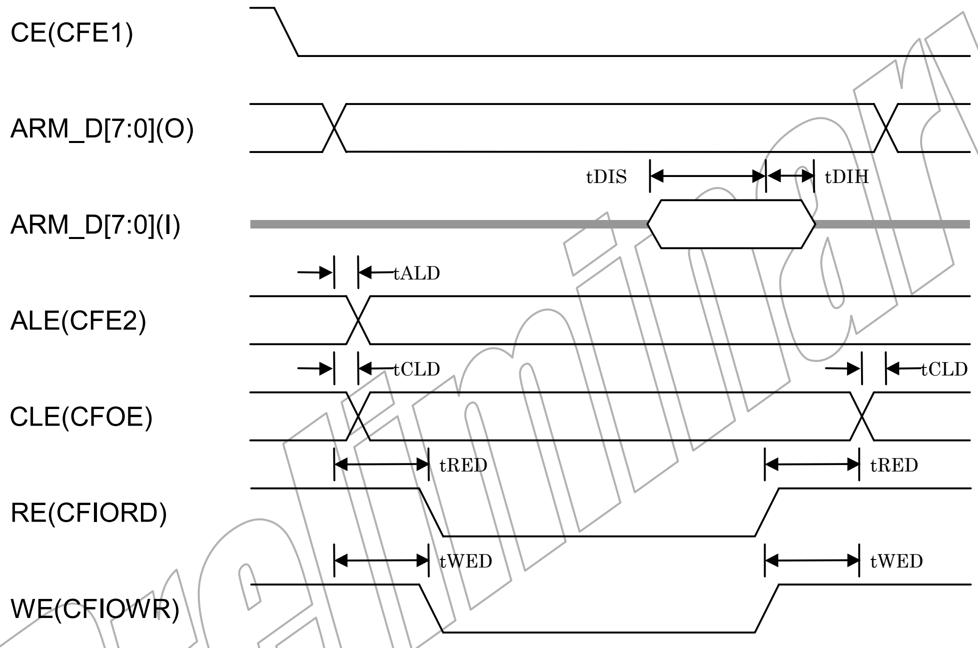


4.8.3 SMART MEDIA Interface

	PARAMETER	MIN	NOM	MAX	UNIT
tALD	ARM_D to ALE delay time	NxTc - 3			ns
tCLD	ARM_D to CLE delay time	NxTc - 2.5		NxTc + 3	ns
tDIS	input data setup time to OE de-asserted	8			ns
tDIH	input data hold time from OE de-asserted	0			ns
tWED	ARM_D to WE delay time	NxTc - 2.5			ns

Tc: ARM clock cycle
N : Programmed number of ARM clocks by specific parameter

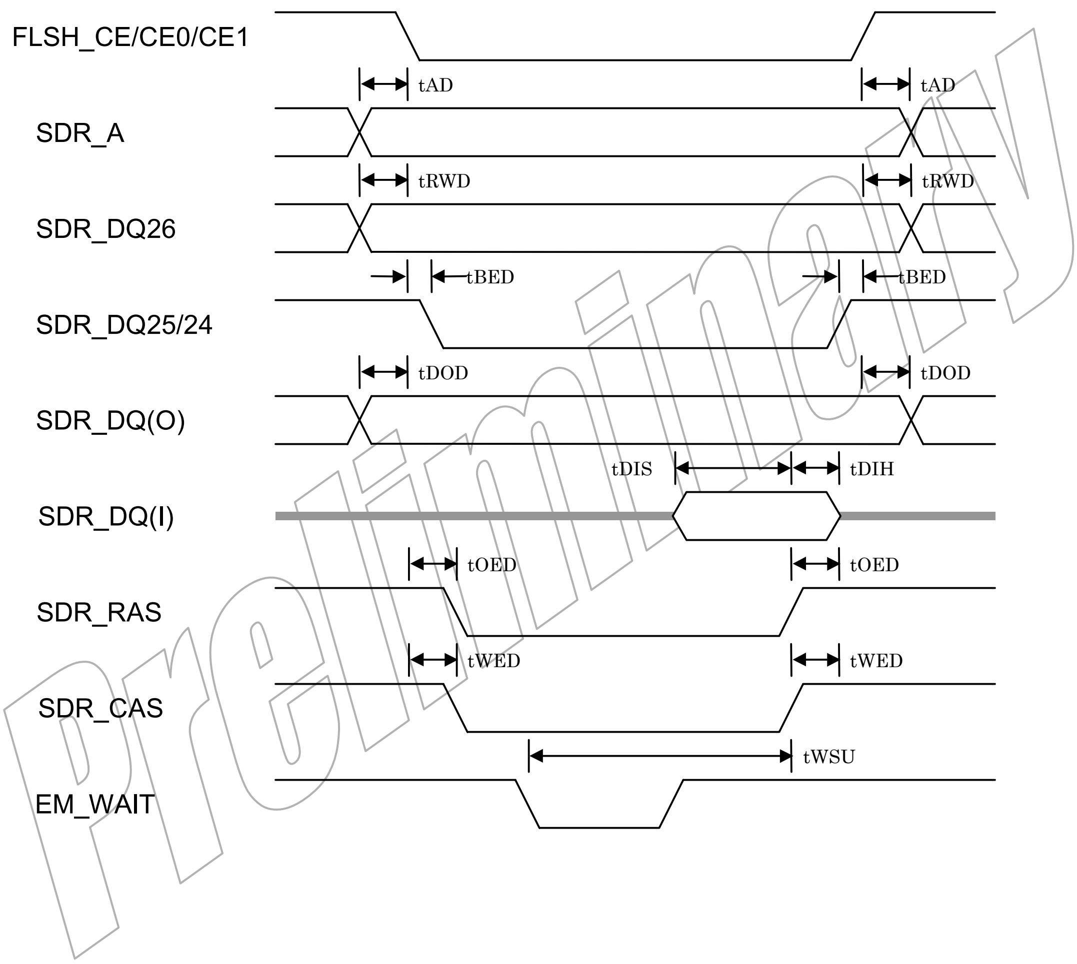
SmartMedia AC timing



4.8.4 FLASH/SRAM Interface (used SDRAM bus)

	PARAMETER	MIN	NOM	MAX	UNIT
tAD	Address to CE delay time	NxTc - 3		NxTc + 7.5	ns
tRWD	RW to CE delay time	NxTc - 3		NxTc + 7	ns
tBED	CE to BE delay time	NxTc - 3		NxTc + 7	ns
tDOD	outut data to CE delay time	NxTc - 9.5		NxTc + 4	ns
tDIS	input data setup time to OE de-asserted	8			ns
tDIH	input data hold time from OE de-asserted	0			ns
tOED	CE to OE delay time	NxTc - 3		NxTc + 6	ns
tWED	CE to WE hold time	NxTc - 3		NxTc + 6	ns
tWSU	WAIT setup time to OE/WE de-aserted	2xTc+6			ns

FLASH&SRAM AC timing

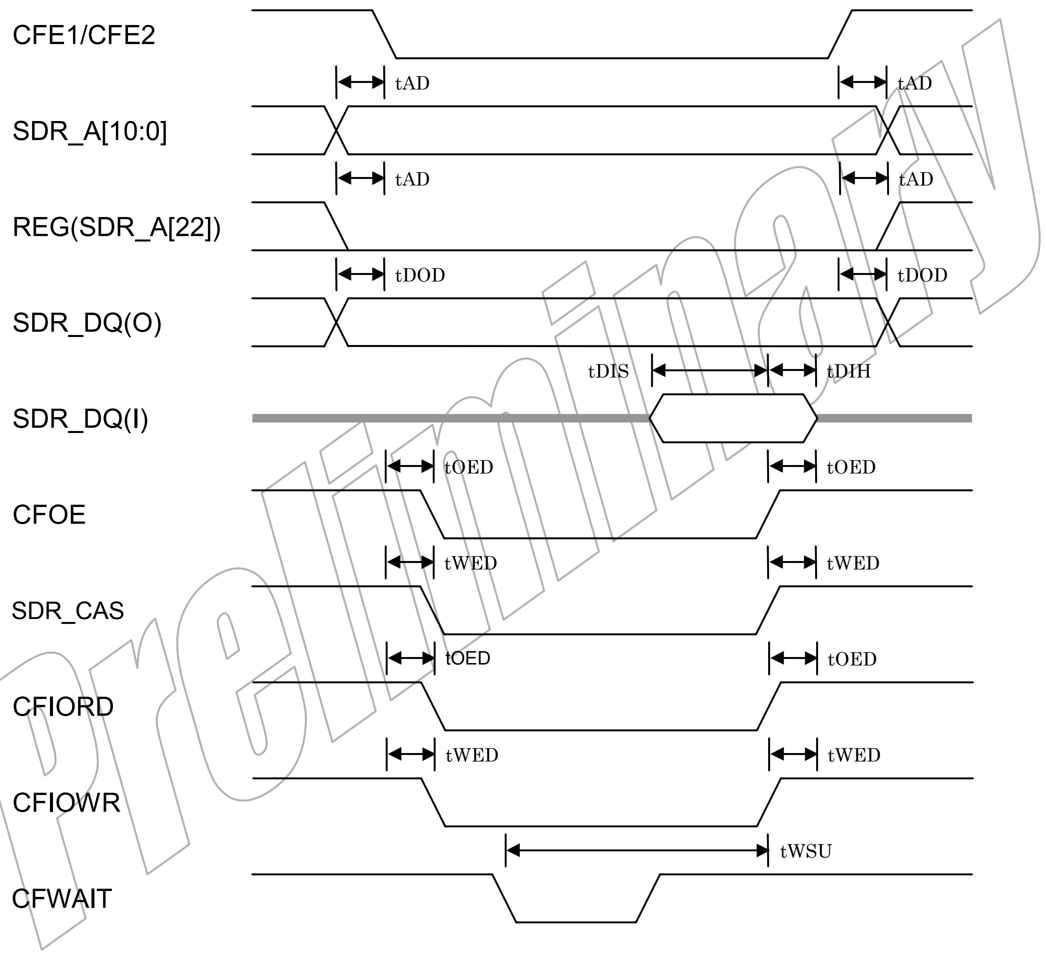


4.8.5 COMPACT FLASH Interface (used SDRAM bus)

	PARAMETER	MIN	NOM	MAX	UNIT
tAD	Address to CE delay time	NxTc - 3		NxTc +7.5	ns
tDOD	outut data to CE delay time	NxTc - 9.5		NxTc + 4	ns
tDIS	input data setup time to OE de-asserted	8			ns
tDIH	input data hold time from OE de-asserted	0			ns
tOED	CE to OE delay time	NxTc - 3		NxTc + 7	ns
tWED	CE to WE hold time	NxTc - 3		NxTc + 5	ns
tWSU	WAIT setup time to OE/WE de-aserted	2xTc+6			ns

Tc: ARM clock cycle
N : Programmed number of ARM clocks by specific parameter

CompactFlash AC timing

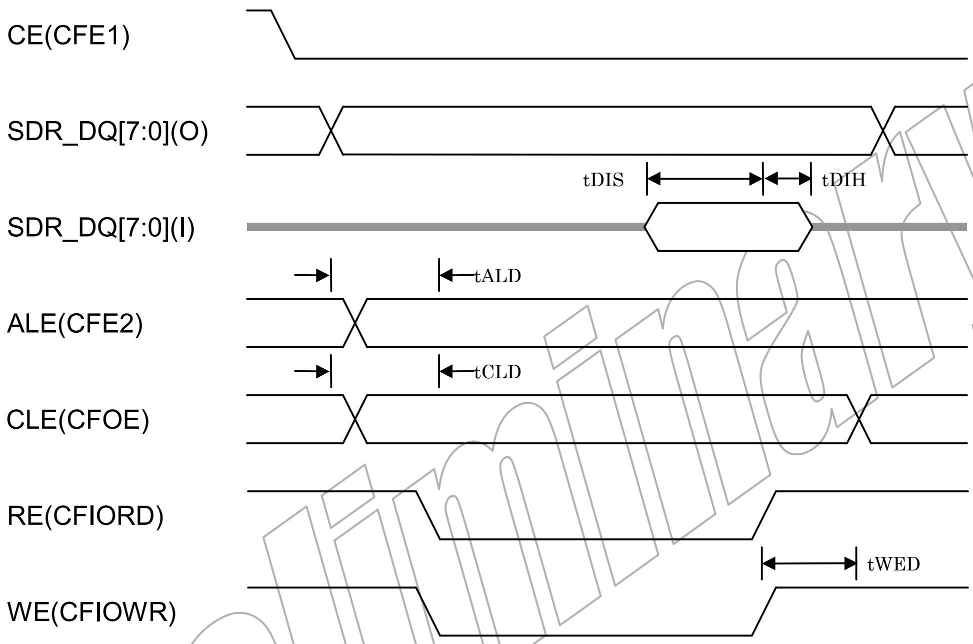


4.8.6 SMART MEDIA Interface (used SDRAM bus)

PARAMETER		MIN	NOM	MAX	UNIT
tALD	ARM_D to ALE delay time	NxTc - 3			ns
tCLD	ARM_D to CLE delay time	NxTc - 2.5		NxTc + 3	ns
tDIS	input data setup time to OE de-asserted	8			ns
tDIH	input data hold time from OE de-asserted	0			ns
tWED	ARM_D to WE delay time	NxTc - 2.5			ns

Tc: ARM clock cycle
N : Programmed number of ARM clocks by specific parameter

<u>SmartMedia AC timing</u>



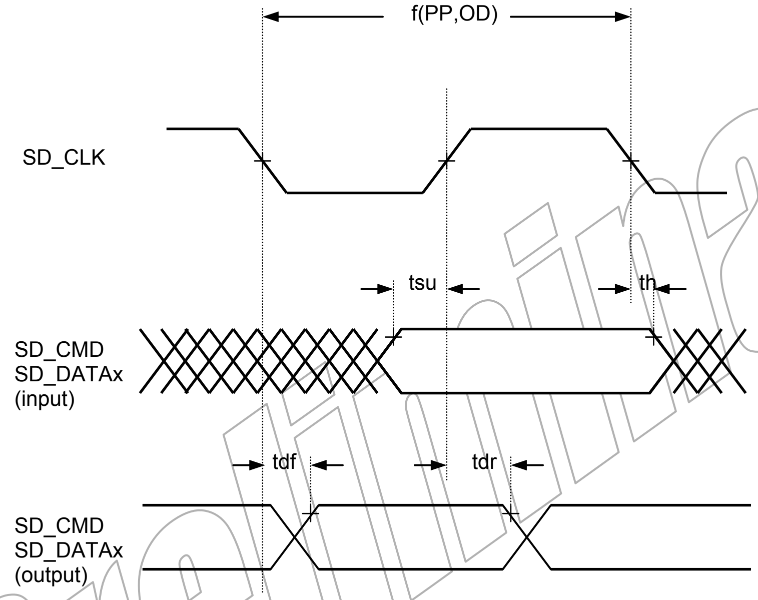
4.9 MultiMedia Card (MMC) Timing

Below table assumes testing over recommended operating conditions.

MultiMedia Card(MMC) Signal Switching Caracteristics

	PARAMETER	MIN	NOM	MAX	UNIT
f(PP)	Clock frequency data transfer mode (PP)	**0**		**20**	MHz
f(OD)	Clock frequency identification mode (OD)	**0**		**400**	kHz
tsu	Input setup time	**10**			ns
th	Input hold time	**0**			ns
tdf	Output delay from SD_CLK falling			**10**	ns
tdr	Output delay from SD_CLK rising	**5**			ns

MultiMedia Card(MMC) Timings



4.10 Secure Digital (SD) Timing

Below table assumes testing over recommended operating conditions.

Secure Digital(SD) Signal Switching Caracteristics

	PARAMETER	MIN	NOM	MAX	UNIT
f(PP)	Clock frequency data transfer mode (PP)	**0**		**20**	MHz
f(OD)	Clock frequency identification mode (OD)	**0**		**400**	kHz
tsu(I)	Input setup time	**10**			ns
th	Input hold time	**0**			ns
tdf	Output delay from SD_CLK falling			**10**	ns
tdr	Output delay from SD_CLK rising	**5**			ns

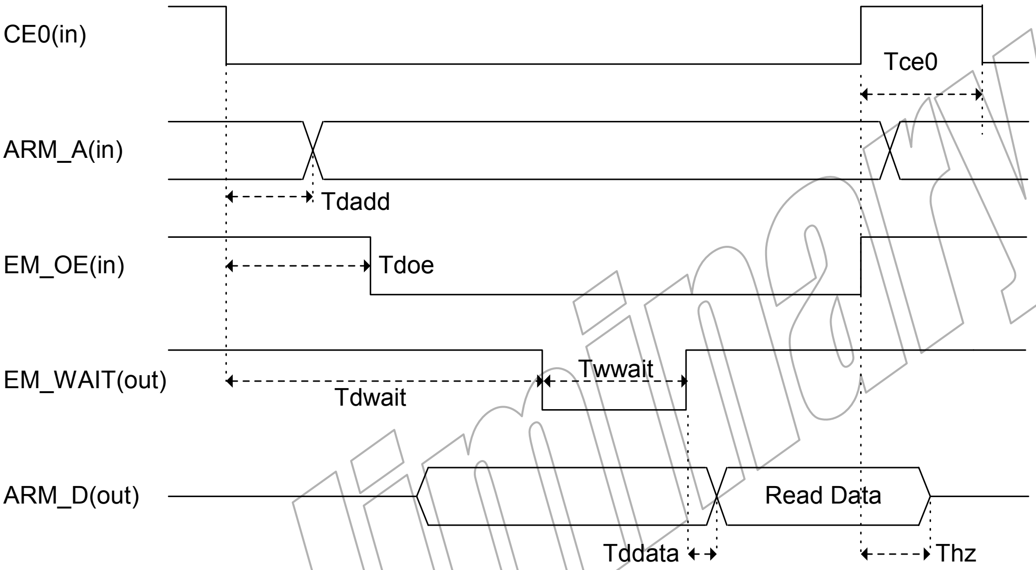
4.11 External Host Interface Timing

External Host Interface Read Timing (DM270 -> External Host)

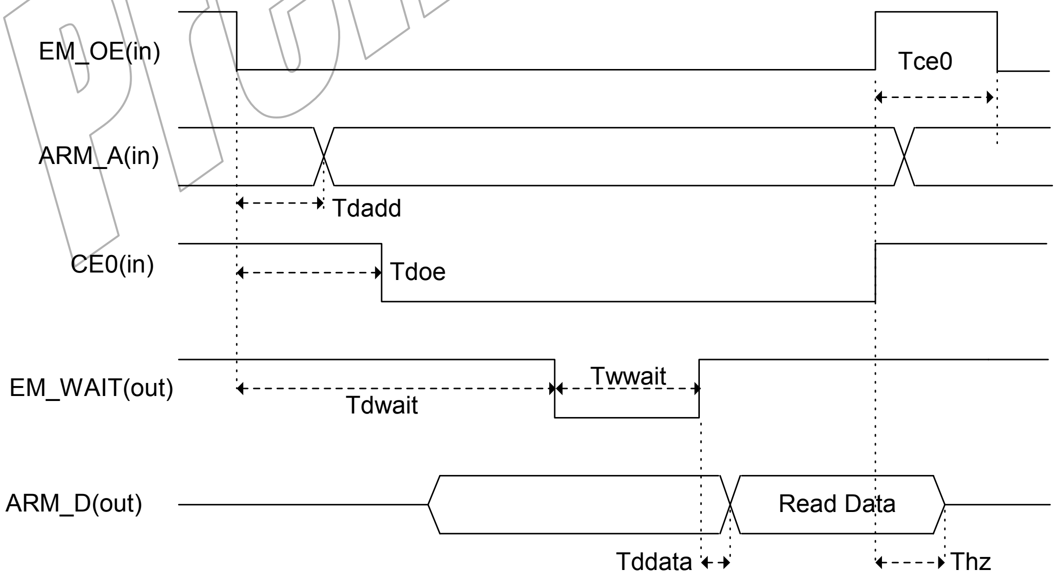
PARAMETER		MIN	NOM	MAX	UNIT
Tdoe	CE to OE input delay time	-		Tcyc - 4	ns
Tdadd	CE to Address input delay time	-		Tcyc - 8	ns
Tdwait	CE to Wait output delay time	-		(Tcyc x 2) + 15	ns
Twwait	Wait width	0		-	ns
Tddata	Data output delay time	-		6.5	ns
Thz	Data output hold time from CE de-asserted	0		12	ns
Tce0	Minimum interval time	1Tcyc + 10		-	ns

Tcyc = DM270 System clock cycle period

External Host Interface Read Timing
(CE trigger mode)



External Host Interface Read Timing
(OE trigger mode)

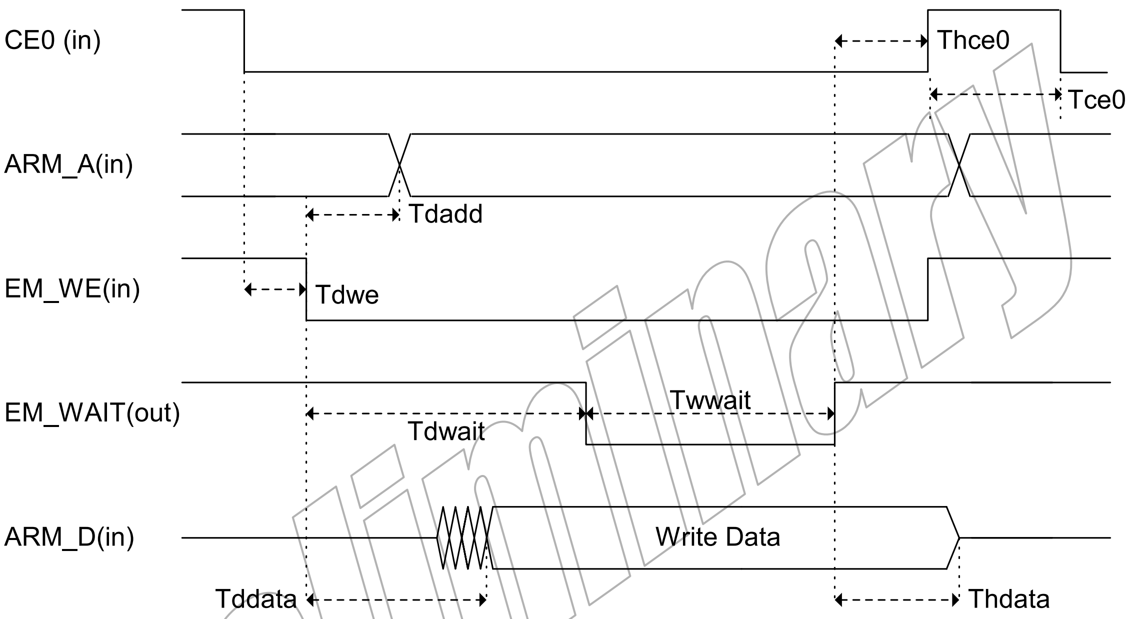


External Host Interface Write Timing (DM270 <- External Host)

PARAMETER		MIN	NOM	MAX	UNIT
Tdwe	CE to WE input delay time	0		-	ns
Tdadd	WE to Address input delay time	-		Tcyc – 8	ns
Tdwait	WE to Wait output delay time	-		Tcycx2 + 14	ns
Twwait	Wait width	0		-	ns
Tddata	WE to data input delay time	-		Tcyc – 0.5	ns
Thce0	CE hold time from Wait de-asserted	0		-	ns
Thdata	Data hold time from Wait de-asserted	0		-	ns
Tce0	Minimum interval time	1Tcyc		-	ns

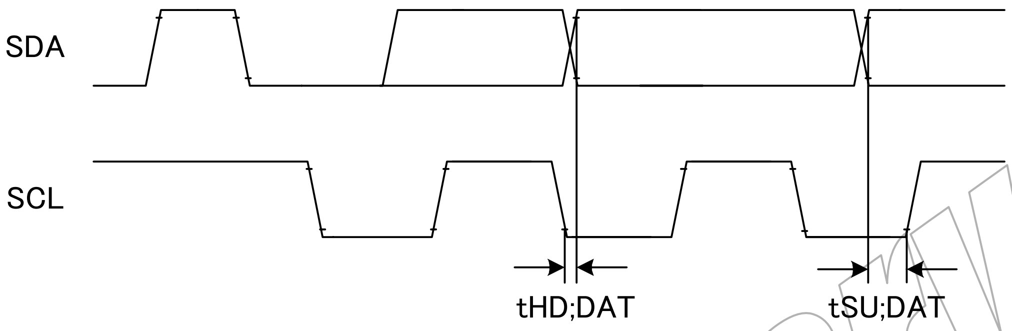
Tcyc = DM270 System clock cycle period

External Host Interface Write Timing



4.12 Pseudo-I2C Bus Timing

	PARAMETER	MIN	NOM	MAX	UNIT
tHD:DAT	Data hold (or delay) time(400kHz)			700	ns
tHD:DAT	Data hold (or delay) time(100kHz)			5100	ns
tSU:DAT	Data set-up time	0		-	ns



5. DAA15 Cell

■ CELL NAME DAA15

■ DESCRIPTION
- 10bit D/A Converter

Signal Name and Function

SIGNAL NAME (*1)	I/O	FUNCTION
BIAS	IN	Bias with phase compensation capacitor
CLK	IN	Master clock
D9 – D0	IN	Data input, D9= MSB, D0= LSB
IREF	IN	Current reference control
VREF	IN	Reference voltage
OE	IN	Output enable control (*2)
PDZ	IN	Power down mode control (*2)
AOUT	OUT	Analog output
TESTDO9-TESTDO0	OUT	Test digital output, TESTDO9=MSB, TESTDO0=LSB

*1) VREF, IREF and BIAS signals are treated as input during the design flow, simulation, and testing. These signals are tied off to high or low through the design flow. On the actual system, these signals are required the reference voltage for VREF pin, the reference current input through dumping resistance for IREF pin, and external compensation capacitance for BIAS pin. These signals are not digital input.
*2) At all mode of DAA15 except for normal mode, AOUT status become Z without load resistor **Raout**. AOUT voltage appears 0(VSSA) level with load resistor **Raout** that is connected to VSSA.
OE=H, PDZ=L : TEST mode, AOUT=0 / TESTDO9-0 activated
OE=H, PDZ=H : Normal mode, AOUT activated / TESTDO9-0=0
OE=L, PDZ=H : Output disable, AOUT=0 / TESTDO9-0=0
OE=L, PDZ=L : Power down mode, AOUT=0 / TESTD9-0=0

Recommended Operating Conditions

	PARAMETER	MIN	TYP	MAX	UNIT
FCLK	Clock input frequency			27	MHz
VO (*1)	Output voltage	VSSA		Vref	V
Vref	Reference voltage range	1.0		1.5	V
Riref	Reference resister		1.6		kΩ
Raout	Output resister		200		Ω
Cbias	Compensation capacitor BIAS to VSSA		0.1		μF

*1) This VO range is kept while the ratio (Riref:Raout) have 8:1 relationship.

Operating Characteristics over Recommended Operating Free-Air Temperature
(V_{DD}=1.5V, V_{DD3V}=3.3V, V_{DDA}=3.3V, V_{ref}=1V, R_{aout}=200Ω, R_{ref}=1.6KΩ)

	PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
ZSET	Zero-scale offset (*1)				10	LSB
FSET	Full-scale offset (*1)				10	LSB
SINL	Static integral non-linearity Error (*1)	f_{clk}=27MHz			±1.0	LSB
SDNL	Static differential non-linearity Error (*1)	f_{clk}=27MHz			±0.5	LSB

*1) These data show the characterization data on EVM.

6. Power Supply Sequencing

It is better that core logic states be stable during power-up/power-down transitions for DVDD. The internal states of inputs and outputs are undefined during the CVDD power-up/power-down transitions and outputs can range from high impedance to full rated source or sink current. Some of the possible states could create bus-contention conditions if other devices are driving the bus. For these reasons, it is preferred design practice to bring up CVDD first and bring it down last.

When there is no probability of bus-contention conditions, the recommended operating conditions listed in the Common Characteristics section of the Definitions Chapter in the Macro Library Summary for the specific product are adequate to protect the device.

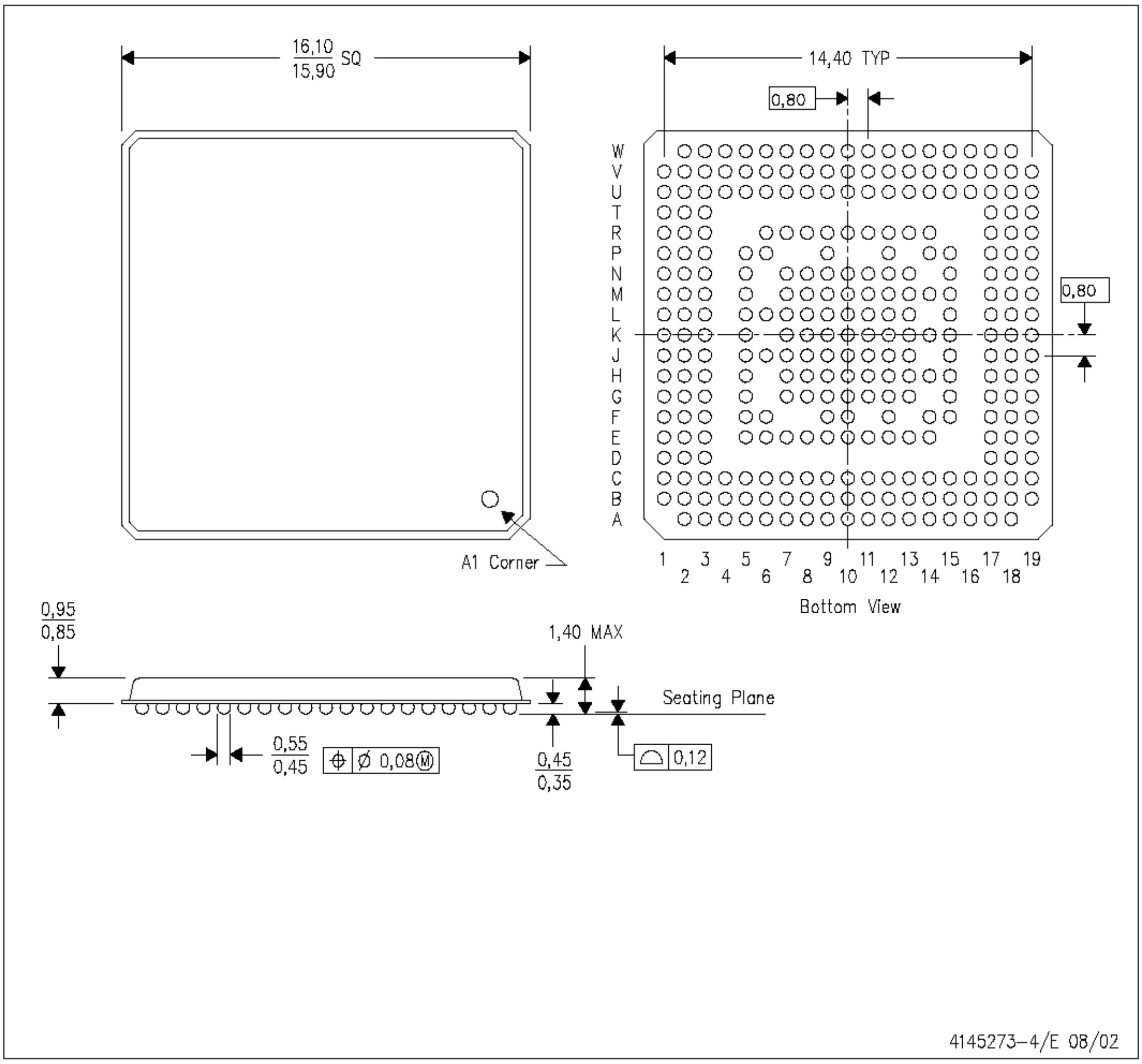
7. Package

7.1 288GHK Size Table

Dimension	W	16.00+/-0.10mm
	D	16.00+/-0.10mm
	H	1.4mm max
Ball Pitch		0.8mm

7.2 288GHK Drawing

GHK (S–PBGA–N288) PLASTIC BALL GRID ARRAY



NOTES:
- A. All linear dimensions are in millimeters.
- B. This drawing is subject to change without notice.
- C. Micro Star BGA configuration.

SCALE	A SIZE	4145273	E REV	SHEET 4 / 5